كلية الحاسبات والذكاء الإصطناعي
Faculty of Computers & Artificial Intelligence

جامعة حلوان
HELWAN UNIVERSITY

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Nextronica

A graduation project dissertation by:

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Supervised by:

Dr. Mohamed Marie

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Finally, for our faculty for providing the suitable environment that leaded us to represent the best image that computer science graduates of Helwan University are supposed to represent.

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Chapter 1

Introduction

1.1 Overview

The project is a web-based e-commerce platform for electronics, built with React and Spring Boot, tailored for Egyptian consumers and local vendors. It offers an extensive bilingual catalog (Arabic and English) across categories like Consumer Electronics, Home Appliances, and Smart Devices. Key features include advanced search, personalized recommendations, real-time inventory tracking.

For vendors, it provides an easy onboarding process, performance analytics, marketing tools, and fair commission structures, empowering small businesses.

The platform's microservices-based, cloud-native architecture ensures scalability, 99.99% uptime, sub-second response times, and robust security, including end-to-end encryption and fraud detection. This project aims to modernize e-commerce in Egypt, enhancing user experience while supporting local innovation.

1.2 Objectives

1. **Develop a Comprehensive Online Marketplace for Electronic Products**

Catalog Scope:

- Extensive range of electronic categories
  - Consumer Electronics
  - Home Appliances
  - Computer Components
  - Smart Home Devices
  - Gaming Accessories

Key Features:

- Sophisticated product classification
- Detailed product information
- Comparison tools
- Bilingual platform (Arabic and English)

2. Provide a Seamless Shopping Experience for Egyptian Consumers

User Experience Goals:

- Intuitive, mobile-responsive design
- Frictionless purchasing process
- Localized payment solutions
- Personalized user interfaces

Technology Enhancements:

- Advanced search capabilities
- Real-time inventory tracking
- Multiple payment methods
- Easy return processes

3. Support Local Electronics Vendors and Businesses

Vendor Empowerment:

- Low-entry barrier for local sellers
- Comprehensive onboarding program
- Fair commission structures
- Marketing and sales support
- Performance analytics tools

Economic Impact:

- Create opportunities for small electronics businesses.
- Reduce digital marketplace entry barriers.

- Showcase local innovation.
- Generate new revenue streams for entrepreneurs.

4. Implement a Robust and Scalable E-Commerce Solution

Technical Architecture:

- Microservices-based infrastructure
- Cloud-native design
- Horizontal scaling capabilities
- Advanced security protocols

Performance Targets:

- 99.99% system uptime
- Sub-second response times
- Robust error handling
- Continuous system optimization

Security Priorities:

- End-to-end encryption
- Compliance with local data regulations
- Advanced fraud detection
- Regular security audits

1.3 Scope

The Nextronica platform will encompass:

- User-friendly web application for customers
- Administrative control panel
- Comprehensive product catalog
- Secure payment and order processing
- Customer support and review mechanisms

1.4 General Constraints

1. *Web-Based Application: Use scalable, responsive, and modular technologies like React and Spring Boot.*

2. *Browser Compatibility: Optimize for modern browsers (Chrome, Firefox, Edge, Safari) with backward compatibility for recent versions.*

3. *Robust Security: Ensure HTTPS, role-based access, encryption, fraud detection, and regular audits.*

4. *Multilingual Support: Provide Arabic (RTL) and English with seamless switching and culturally relevant translations.*

5. *Regulatory Compliance: Adhere to Egyptian e-commerce laws, including data privacy, VAT, approved payment gateways, and clear return policies.*

Chapter 2

Project "Planning & Analysis"

2.1 Project Planning

2.1.1 Feasibility Study

Problem Statement

The Egyptian electronics market faces significant challenges in providing accessible, reliable,
and comprehensive online shopping experiences. Existing platforms suffer from:

- Limited product variety
- Lack of support for local vendors
- Inefficient purchasing processes
- Inadequate customer support
- Limited digital marketplace options for Egyptian consumers

Nextronica aims to address these challenges by creating a comprehensive, user-friendly e-commerce platform that serves the unique needs of the Egyptian electronics market.

Cost Analysis

Initial Development Costs

1. Software Development

- Backend Development: 150,000 EGP
- Frontend Development: 100,000 EGP
- Mobile App Development: 80,000 EGP
- Total Software Development: 330,000 EGP

2. Infrastructure

- Cloud Hosting (Annual): 60,000 EGP
- Server Setup: 40,000 EGP
- Network Security: 25,000 EGP
- Total Infrastructure: 125,000 EGP

3. Design and User Experience

- UI/UX Design: 50,000 EGP
- Branding: 30,000 EGP
- Total Design Costs: 80,000 EGP

4. Marketing and Launch

- Initial Marketing Campaign: 100,000 EGP
- Social Media Advertising: 50,000 EGP
- Content Creation: 30,000 EGP
- Total Marketing: 180,000 EGP

Operational Costs (Annual)

1. Human Resources

- Technical Team Salaries: 600,000 EGP
- Customer Support: 240,000 EGP
- Management: 300,000 EGP
- Total HR Costs: 1,140,000 EGP

2. Ongoing Technical Expenses

- Cloud Hosting: 72,000 EGP
- Software Maintenance: 100,000 EGP
- Security Updates: 50,000 EGP
- Total Technical Expenses: 222,000 EGP

Revenue Projection

5. Revenue Streams

- Vendor Commission (15%): Estimated 500,000 EGP/year
- Advertising Revenue: 100,000 EGP/year
- Total Estimated Revenue: 600,000 EGP/year

Cost Summary

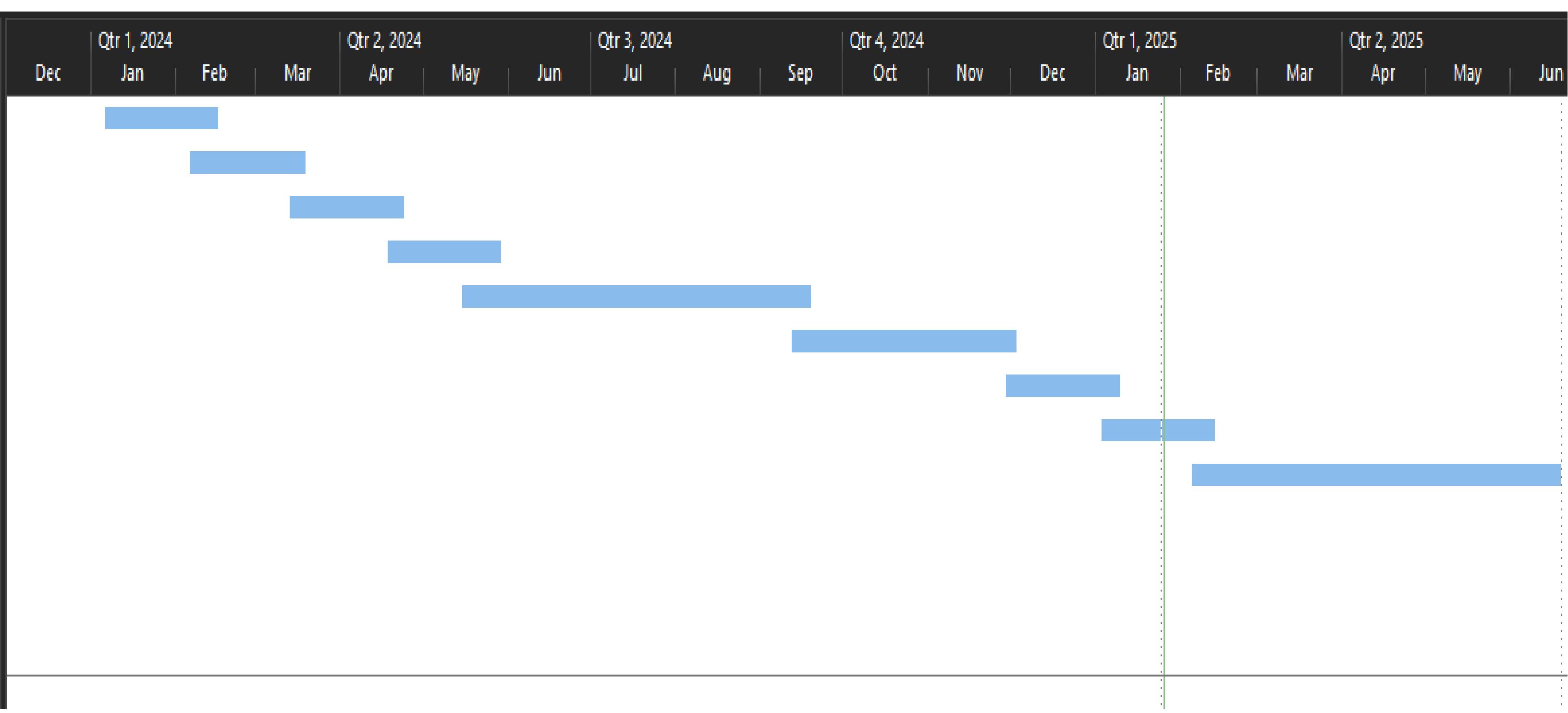
- Total Initial Investment: 715,000 EGP
- Annual Operational Costs: 1,362,000 EGP
- Estimated Annual Revenue: 750,000 EGP

Cost Benefits

- Reduced overhead compared to physical retail.
- No warehouse maintenance costs.
- Minimal physical infrastructure requirements
- Scalable business model
- Low entry barrier for vendors

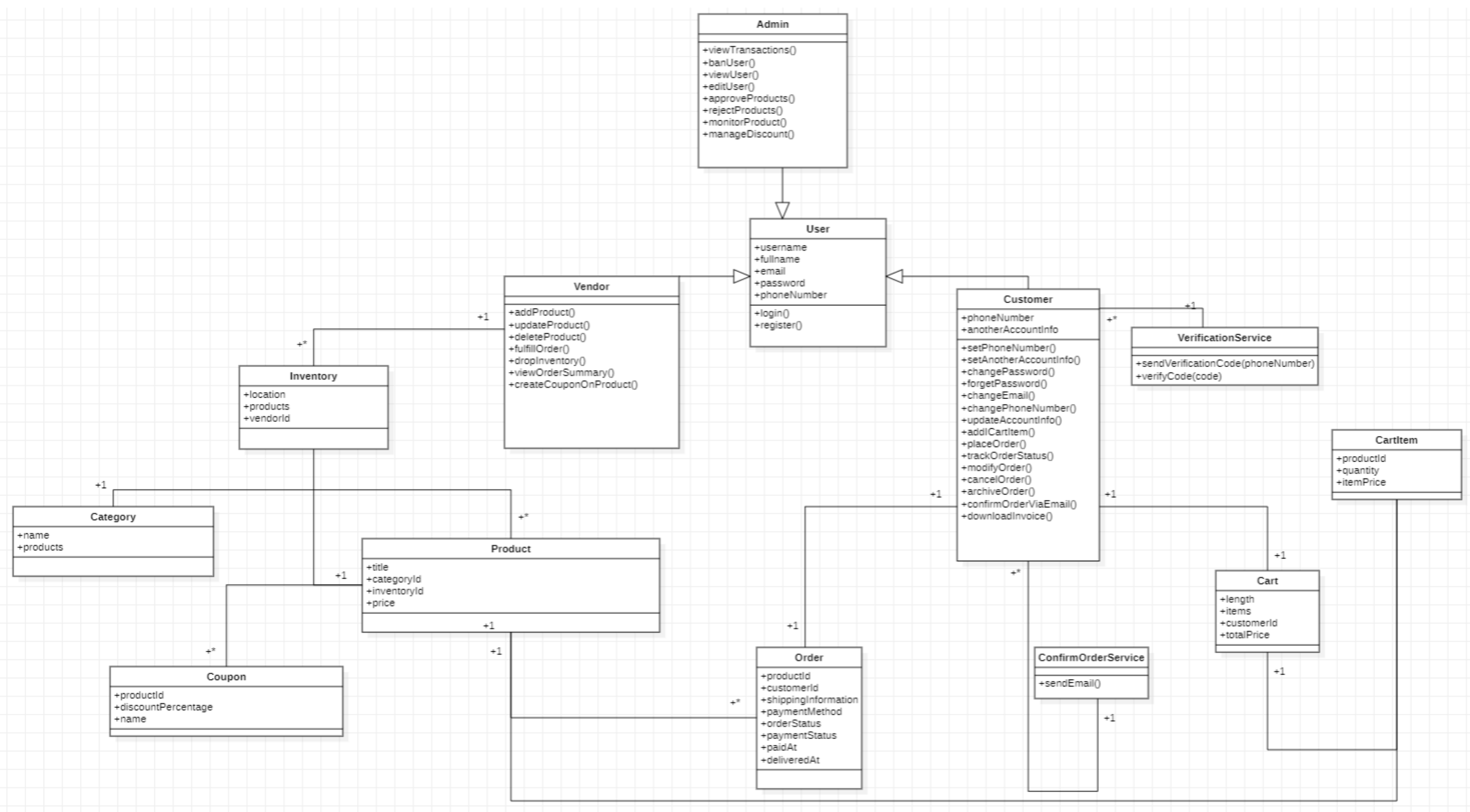
2.1.2 Gantt Chart

Task	Start Date	End Date	Duration
Planning and Reuirements Gathering	6/1/2024	7/1/2024	30
Design	7/1/2024	7/31/2024	30
User Interface Development	8/1/2024	8/31/2024	30
Diagrams	9/1/2024	9/30/2024	29
Backend Development	10/1/2024	12/31/2024	91
Database and Service Integration	1/1/2025	2/28/2025	58
Testing Phase	3/1/2025	3/31/2025	30
Deployment Preparation	4/1/2025	4/30/2025	29
Launch and Intial Support	5/1/2025	7/31/2025	91



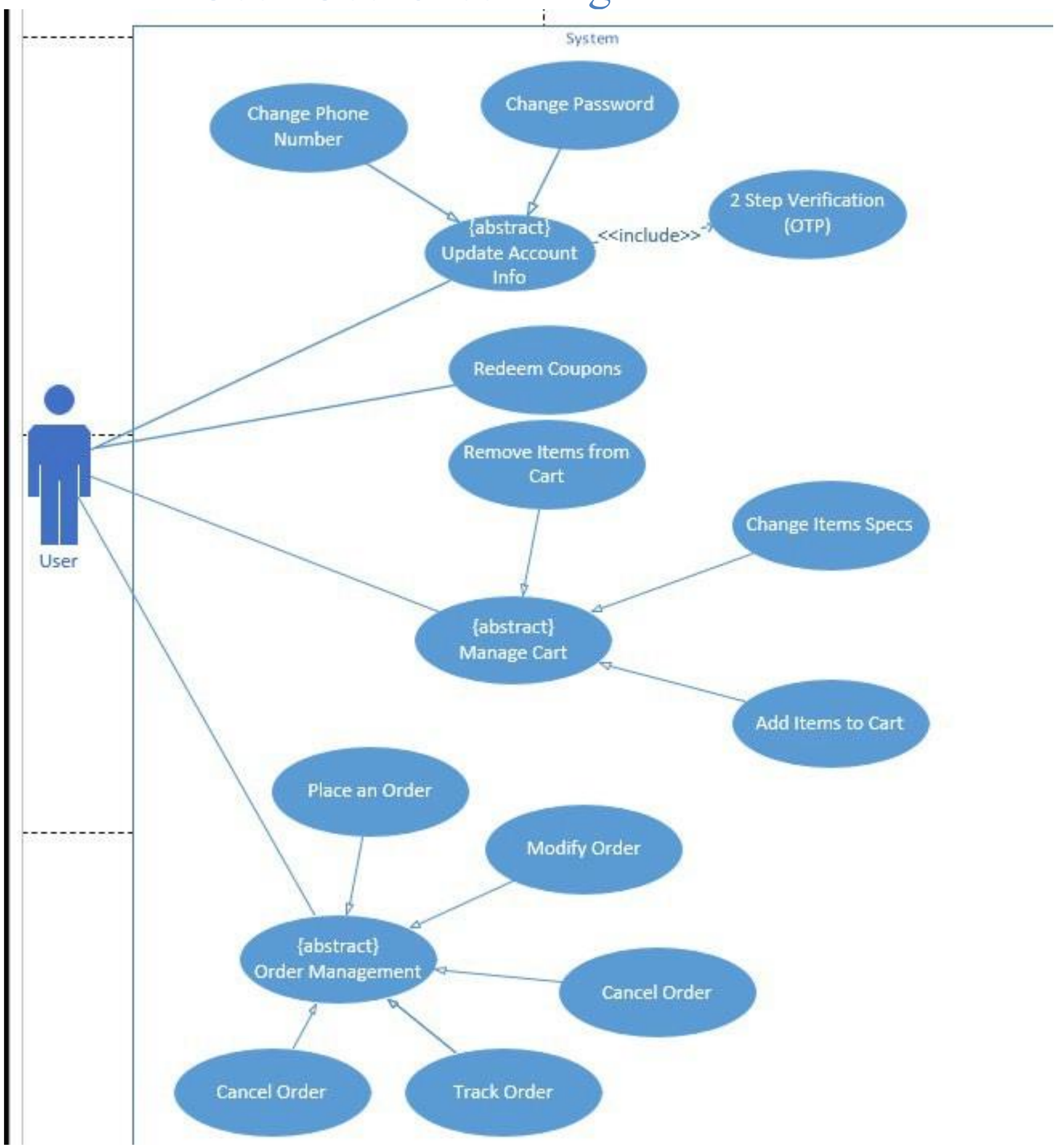
2.1.3 System Design & Functional Requirements

1.1 Class Diagram

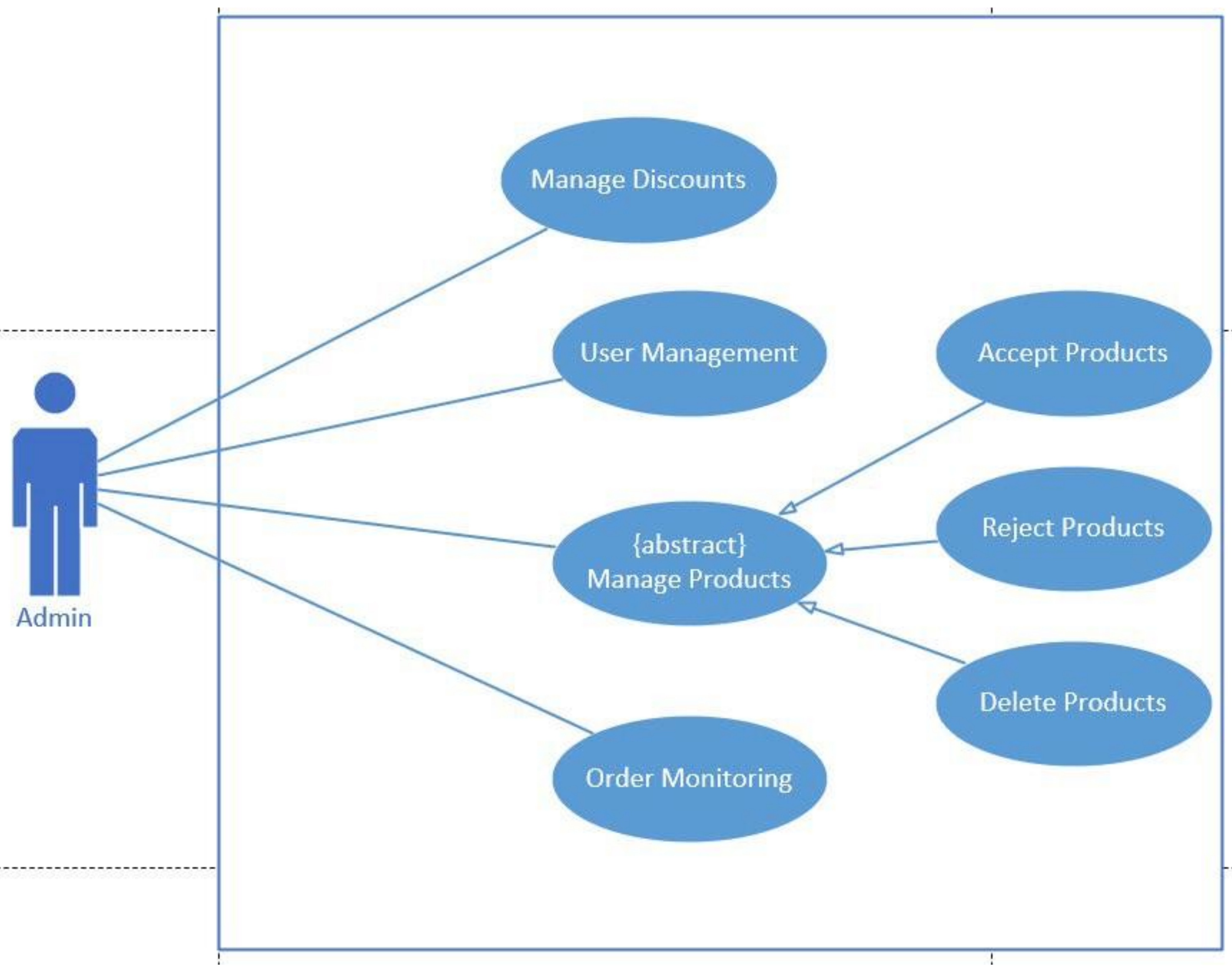


1.2 Use Case Diagrams

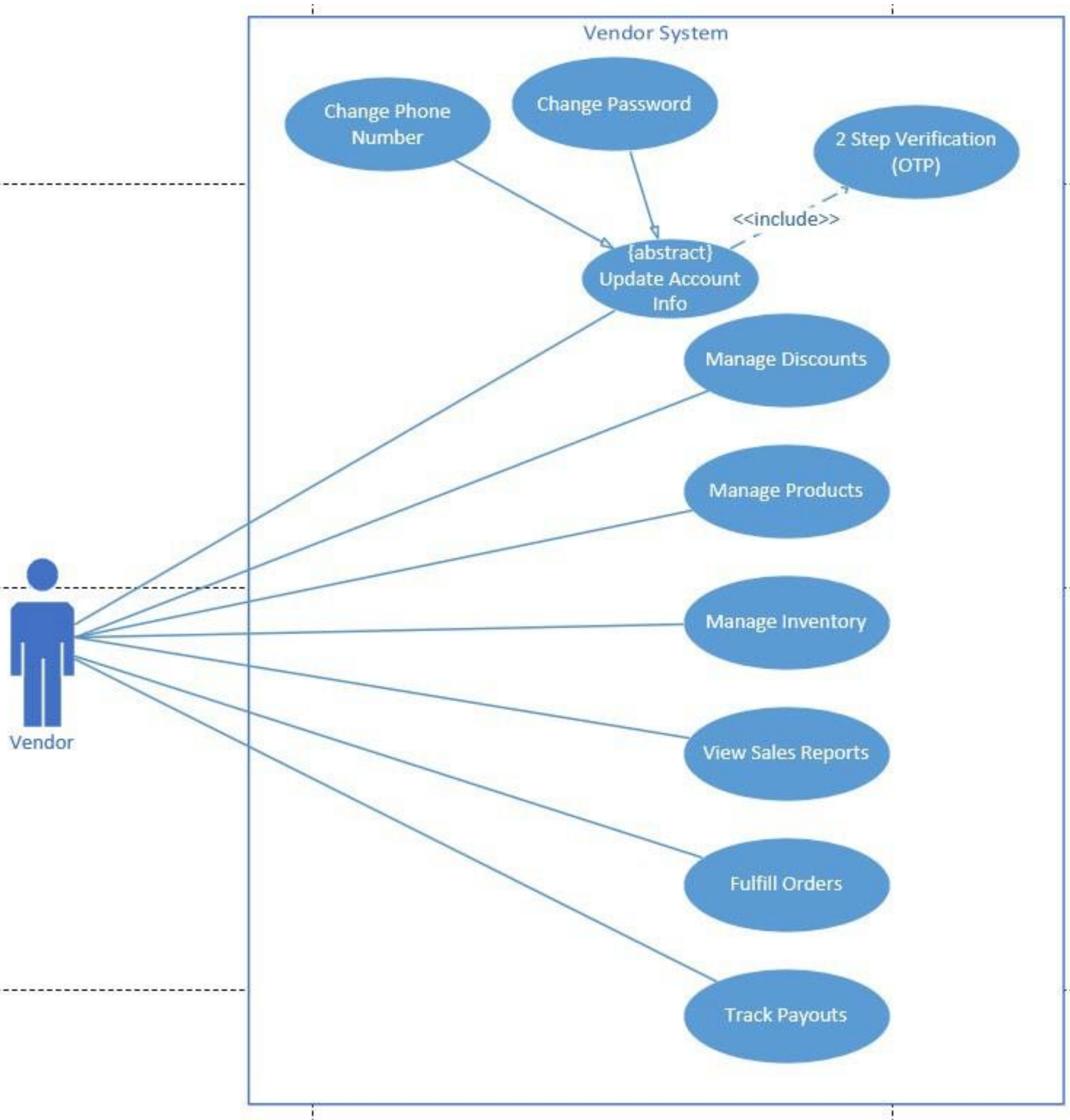
User Use Case Diagram



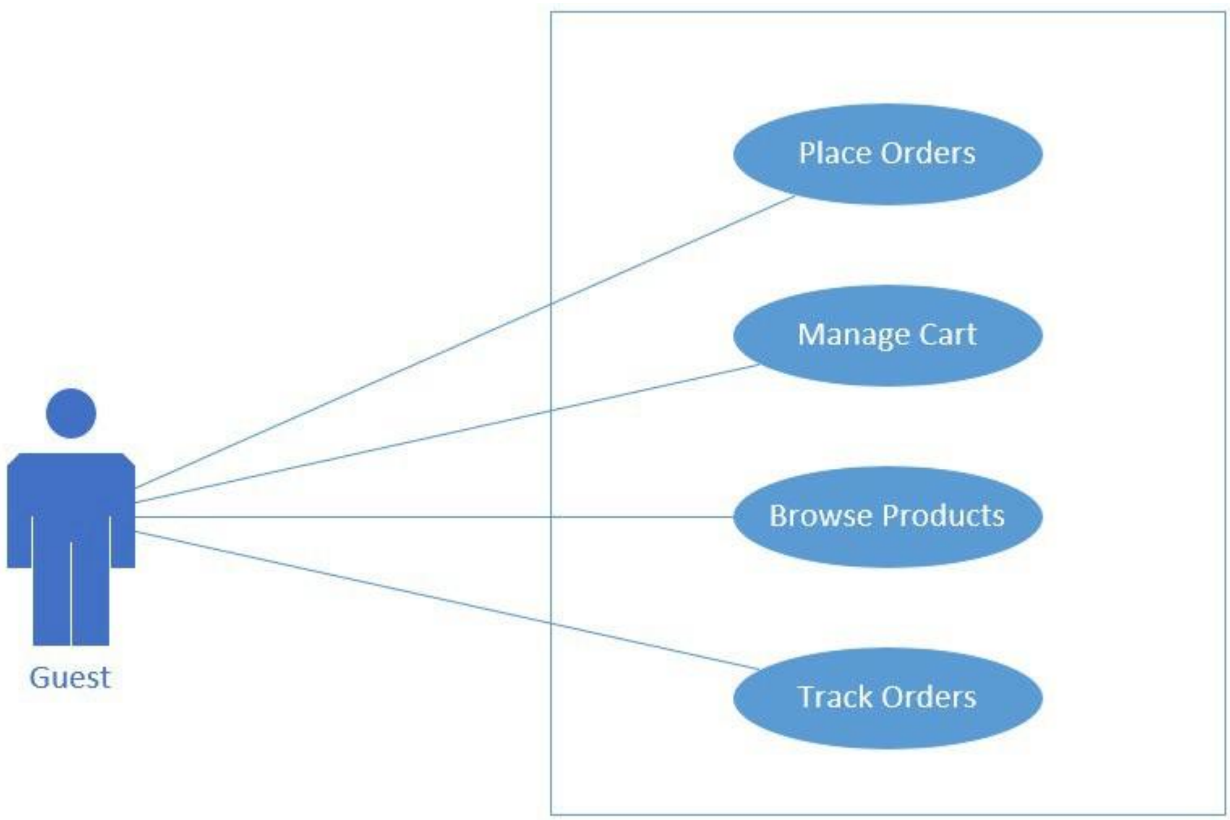
Admin Use Case Diagram



Vendor Use Case Diagram



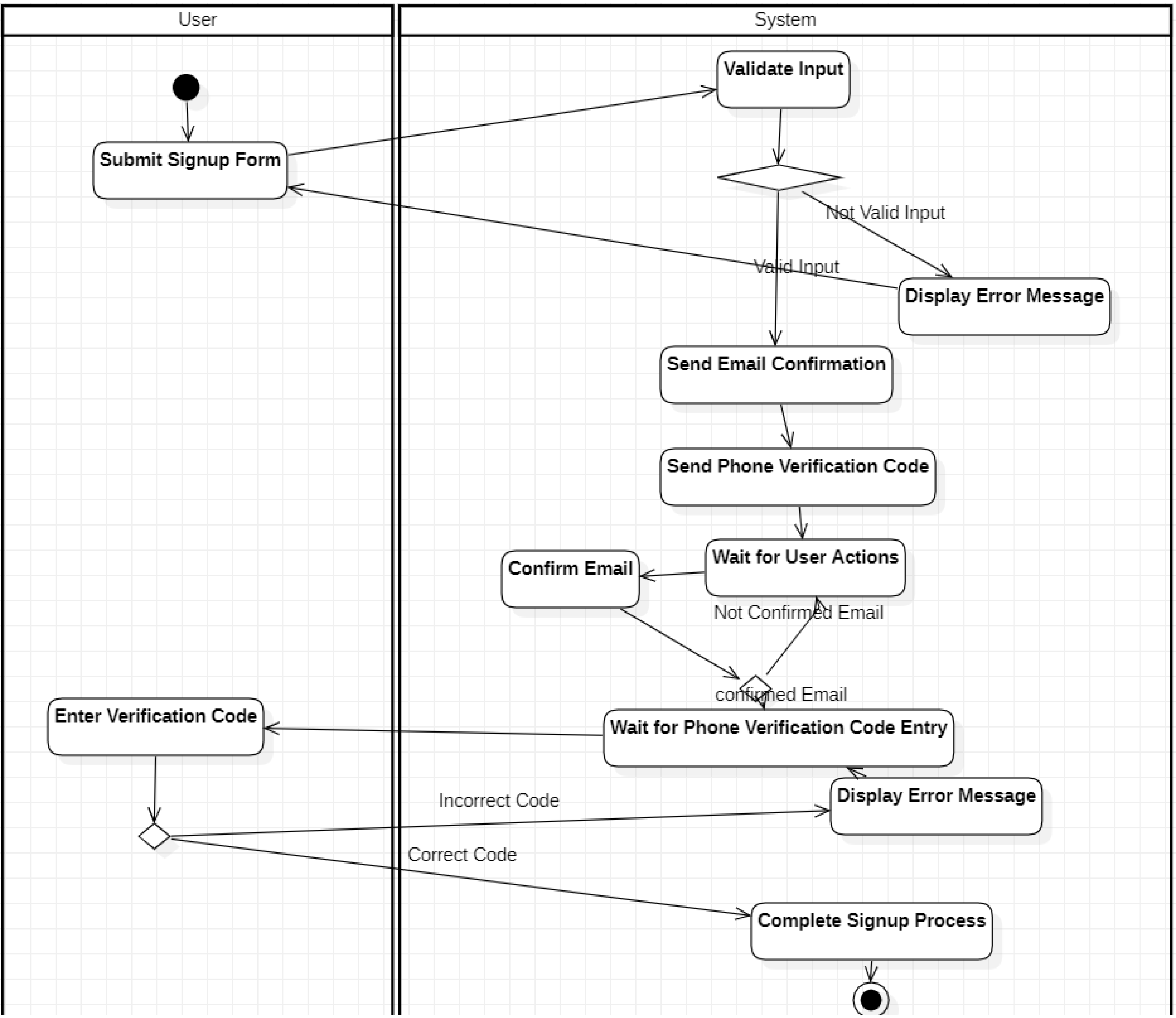
Guest Use Case Diagram



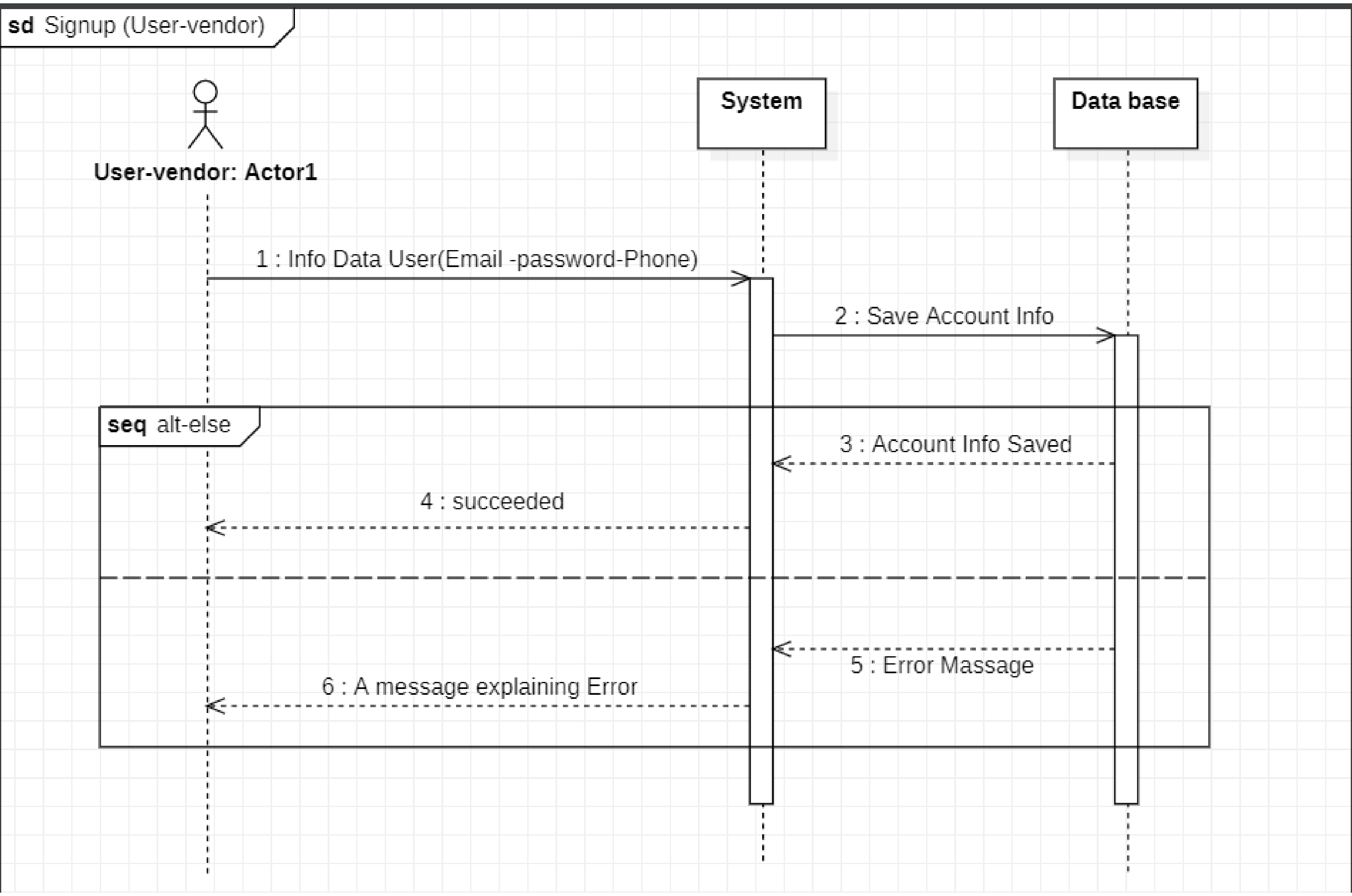
1.1 Sign Up

Use Case Title	Signup with Email Confirmation and Phone Number Verification	
Use Case ID	UC-001	
Actor	A new user who wants to register for the service	
Goal	To successfully register a new user by collecting their email and phone number, confirming the email address, and verifying the phone number.	
Pre-conditions	1- The user must have access to a valid email address and phone number. 2- The user must have internet access. 3- The signup page must be accessible and functioning.	
Post-conditions	1- The user is successfully registered and has an active account. 2-The email and phone number are verified. 3-The user receives confirmation notifications for both the email and phone number.	
Exceptions	1- If the email is already registered, display an error message. 2-If the phone number is already verified or registered, display an error message. 3-If the phone number verification code is incorrect or expired, prompt the user to request a new code.	
Main Flow	User 1- **User Accesses Signup Page**. **2- User Enters Details**. **3- Submit Signup Form**. **4- User Checks Email**. **5**- **User Enters Phone Verification Code** **6- Signup Complete**.	System **1- System Displays Signup Page**. **2-System Validates User Information**. **3-System Sends Email Confirmation**. **4- System Sends Phone Verification Code**. **5- System Verifies Email Confirmation.** **6- System Verifies Phone Verification Code.** **7-System Redirects User to Welcome Page.**

Activity Diagram



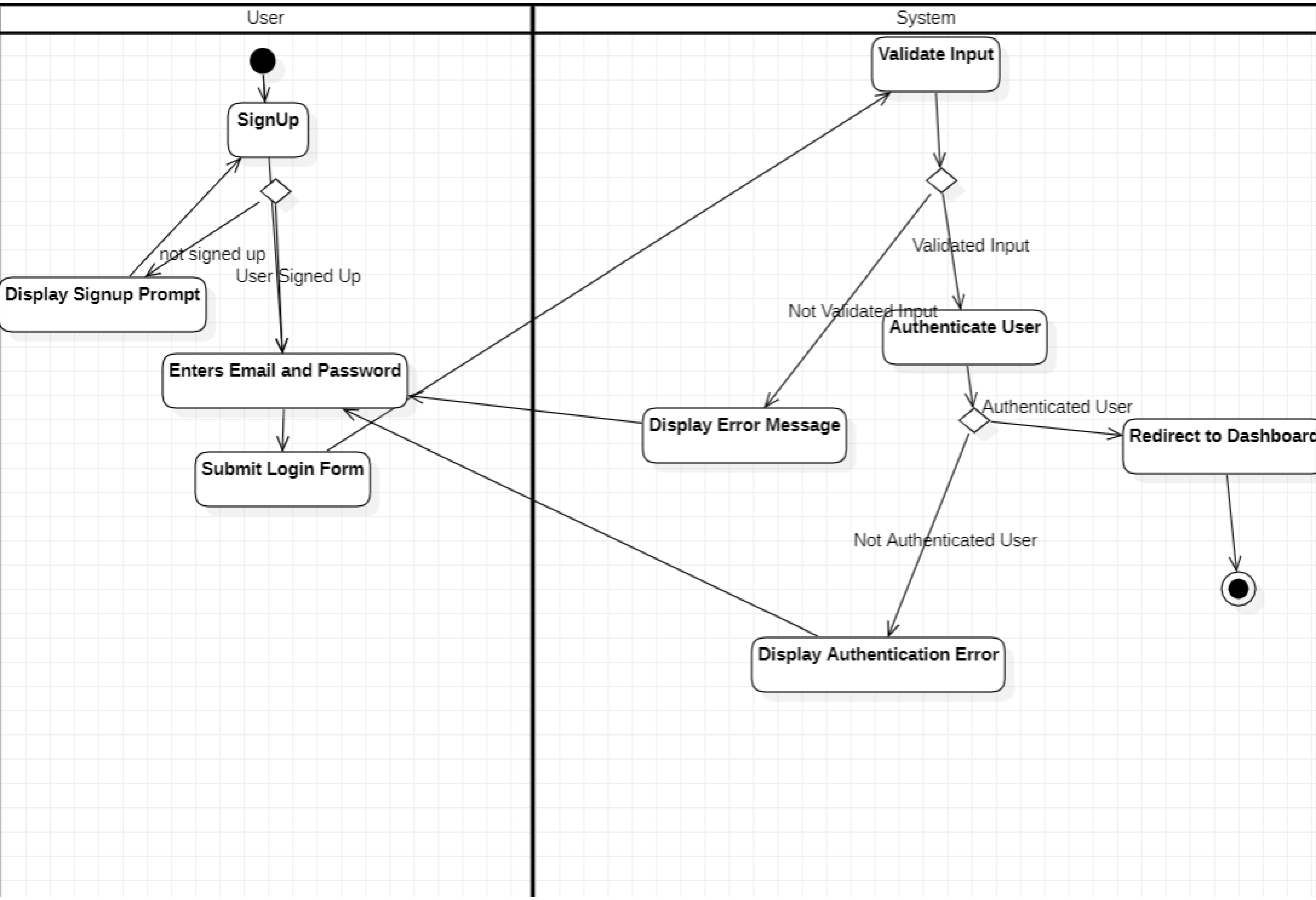
Sequence Diagram



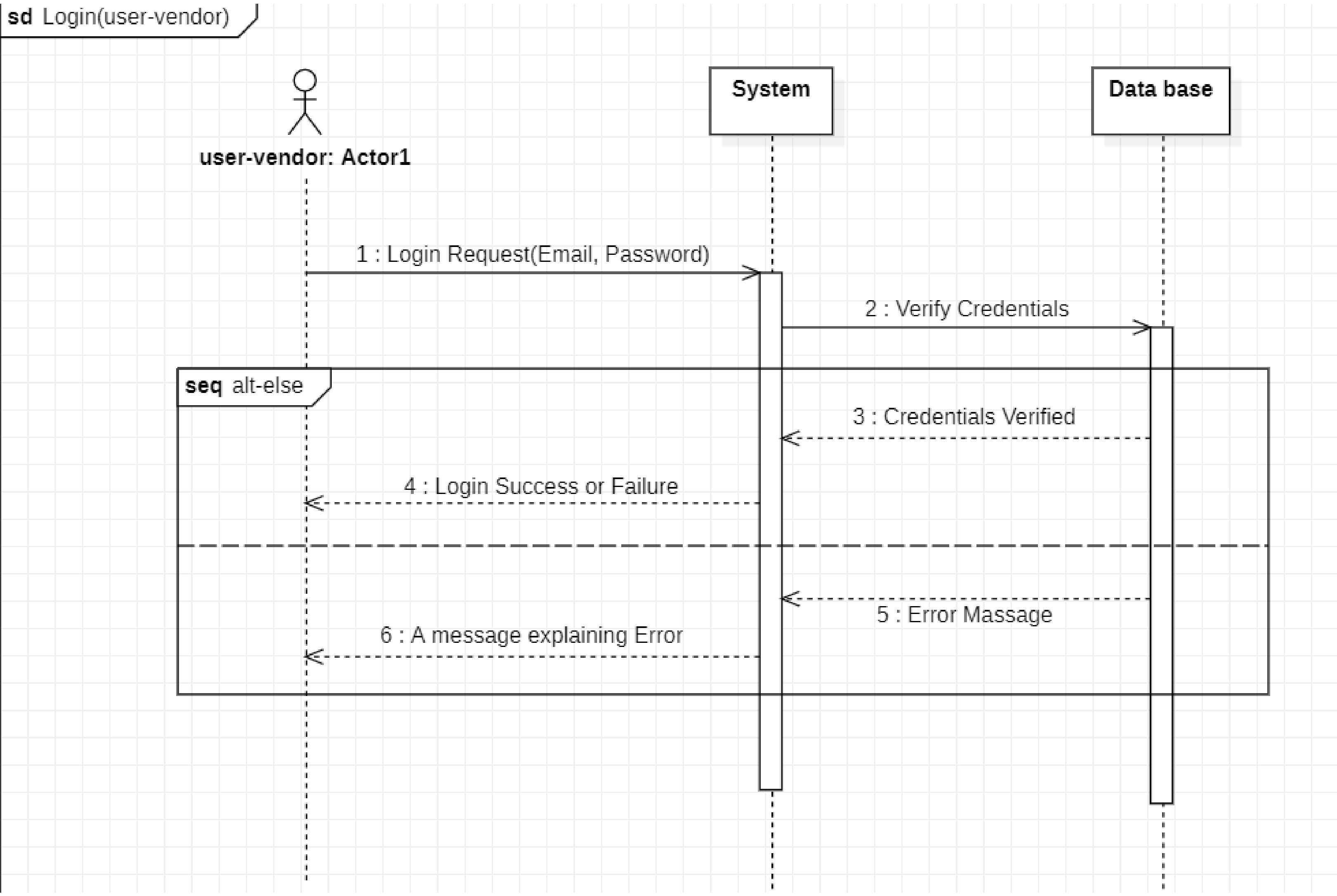
1.2 Login

<table>
<tr><td>Use Case Title</td><td colspan="2">Login</td></tr>
<tr><td>Use Case ID</td><td colspan="2">UC-002</td></tr>
<tr><td>Actor</td><td colspan="2">User: An existing user who wants to access their account.</td></tr>
<tr><td>Goal</td><td colspan="2">To allow an existing user to securely log in to their account using their email and password.</td></tr>
<tr><td></td><td colspan="2"></td></tr>
<tr><td>Pre-conditions</td><td colspan="2">The user must have a registered account with a valid email and password.</td></tr>
<tr><td>Post-conditions</td><td colspan="2">1-The user is successfully logged in and redirected to their account dashboard.
2-The user session is initiated for the duration of the login.</td></tr>
<tr><td>Exceptions</td><td colspan="2">1-If the email is not registered, display an error message.
2-If the password is incorrect, display an error message.
3-If the account is locked or suspended, display an error message.</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">*Alternative Flow for Incorrect Password, Email:
4A. The system detects that the Credentials are incorrect.
4B. The system displays an error message and allows the user to retry.
*Alternative Flow for Locked Account:
4C. The system detects that the account is locked or suspended.
4D. The system displays an error message informing the user of the account status.</td></tr>
<tr><td>Main Flow</td><td>User
1-User Accesses Login Page.
2-User Enters Credentials.
3-Submit Login Form.
4-Successful Login.
5-Login Complete.</td><td>System
1-System Displays Login Page.
2-System Validates Credentials.
3-System Creates User Session.
4-System Redirects User.
5-System Displays Welcome Dashboard</td></tr>
</table>

Activity Diagram



Sequence Diagram



1.3 Change Password

<table>
<tr><td>Use Case Title</td><td colspan="2">Change Password</td></tr>
<tr><td>Use Case ID</td><td colspan="2">UC-003</td></tr>
<tr><td>Actor</td><td colspan="2">User: An authenticated user who wants to change their password.</td></tr>
<tr><td>Goal</td><td colspan="2">To enable the user to securely change their account password.</td></tr>
<tr><td>Pre-conditions</td><td colspan="2">1-The user must be logged into their account.
2-The user must know their current password.
3-The user must have access to the password change page.</td></tr>
<tr><td>Post-conditions</td><td colspan="2">1-The user’s password is successfully updated.
2-The user can log in with the new password.</td></tr>
<tr><td>Exceptions</td><td colspan="2">1-If the current password entered is incorrect, display an error message.
2-If the new password does not meet security requirements (e.g., length, complexity), display an error message.
3-If there’s a system error while processing the password change, notify the user.</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">*Alternative Flow for Incorrect Current Password:
• 5A. The system detects that the current password is incorrect.
• 5B. The system displays an error message and allows the user to retry.
*Alternative Flow for Invalid New Password:
• 6A. The system detects that the new password does not meet security requirements.
• 6B. The system displays an error message explaining the requirements.</td></tr>
<tr><td>Main Flow</td><td>User
1-User Accesses Change Password Page.

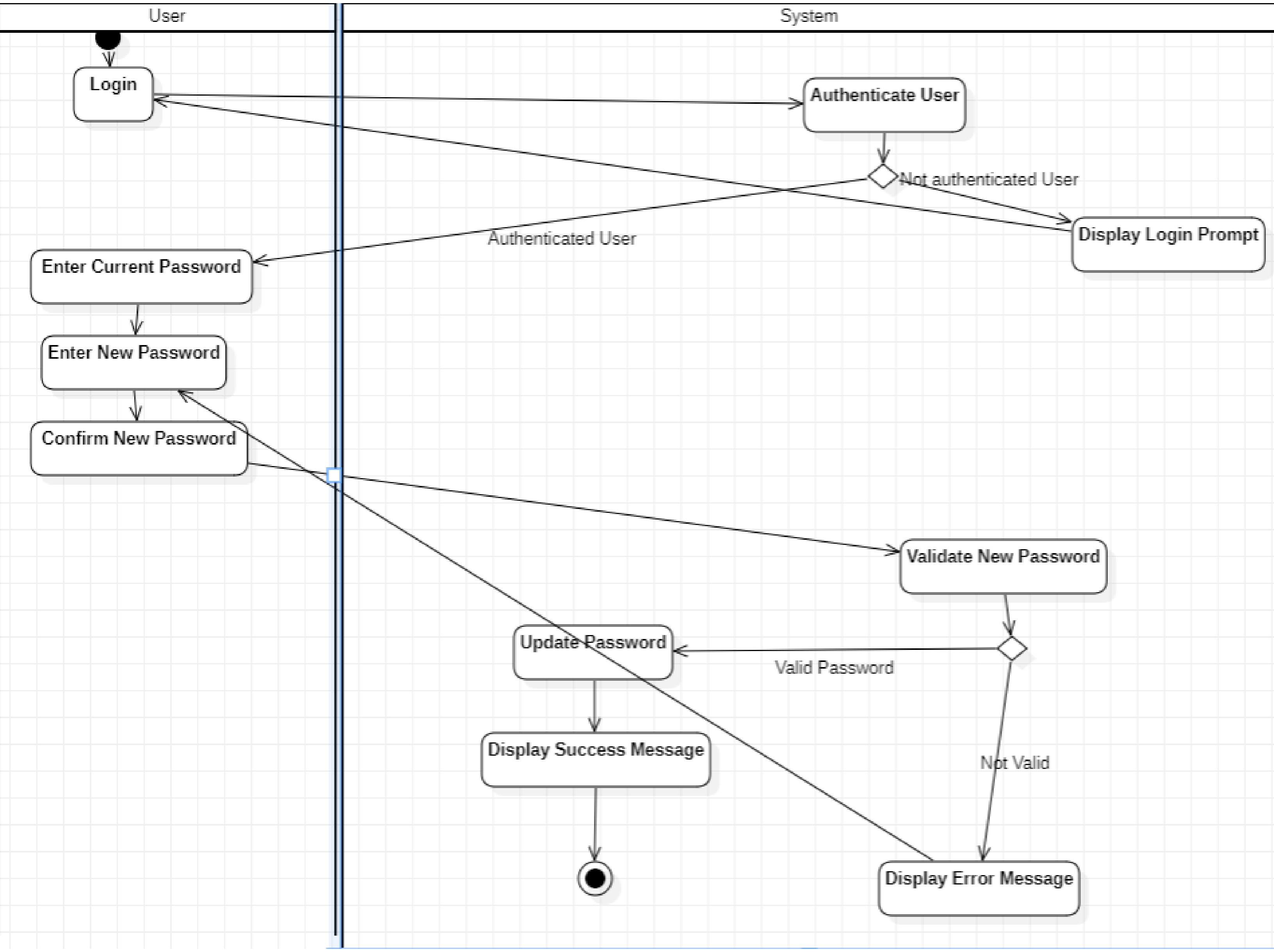
2- User Enters Current Password.

3-User Enters New Password.

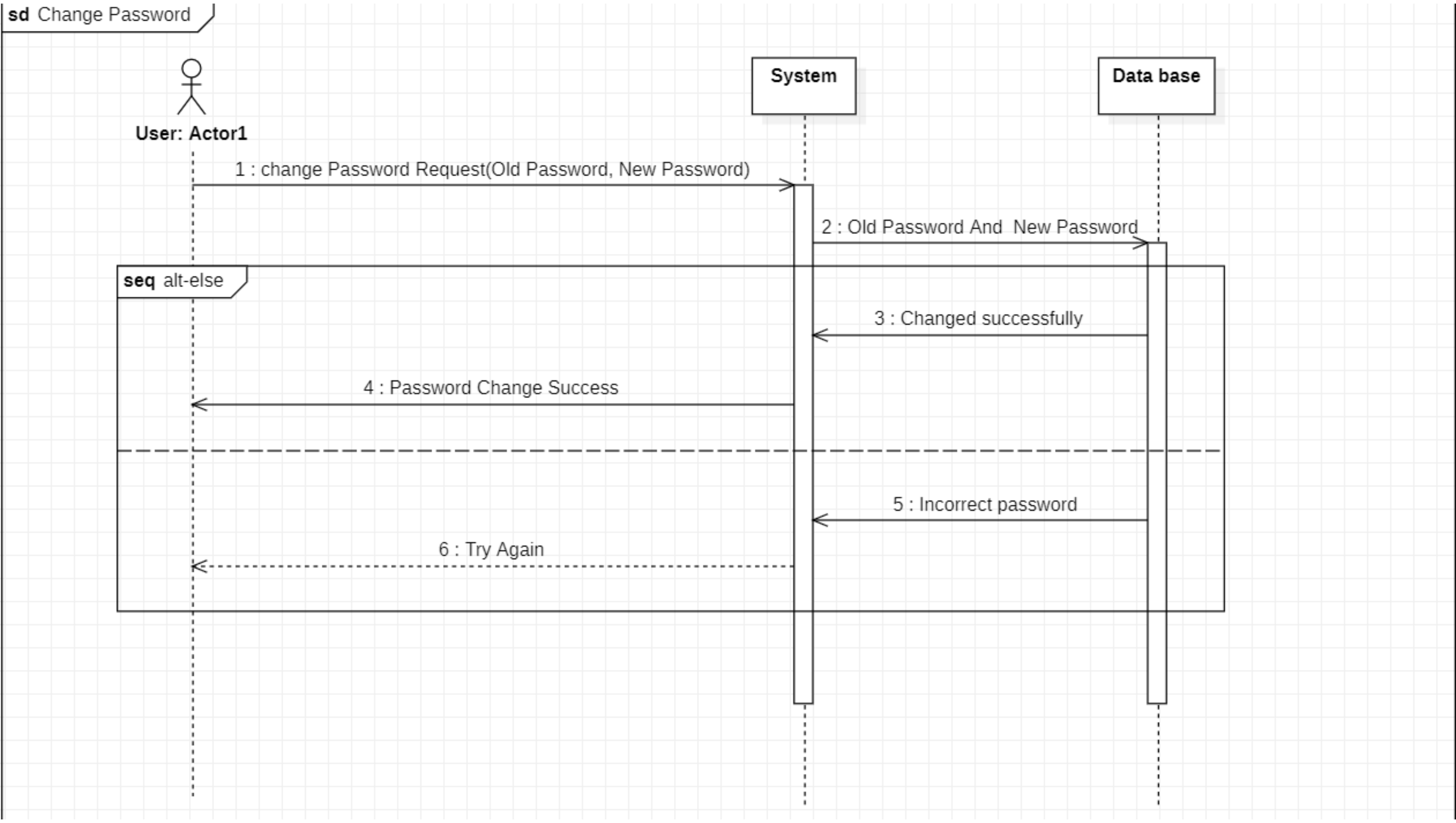
4-Submit Password Change Form:

5-Logout Prompt.</td><td>System
1-System Displays Change Password Page:
2-System Validates Current Password.
3-System Validates New Password.
4-System Updates Password.
5-System Sends Confirmation.</td></tr>
</table>

Activity Diagram



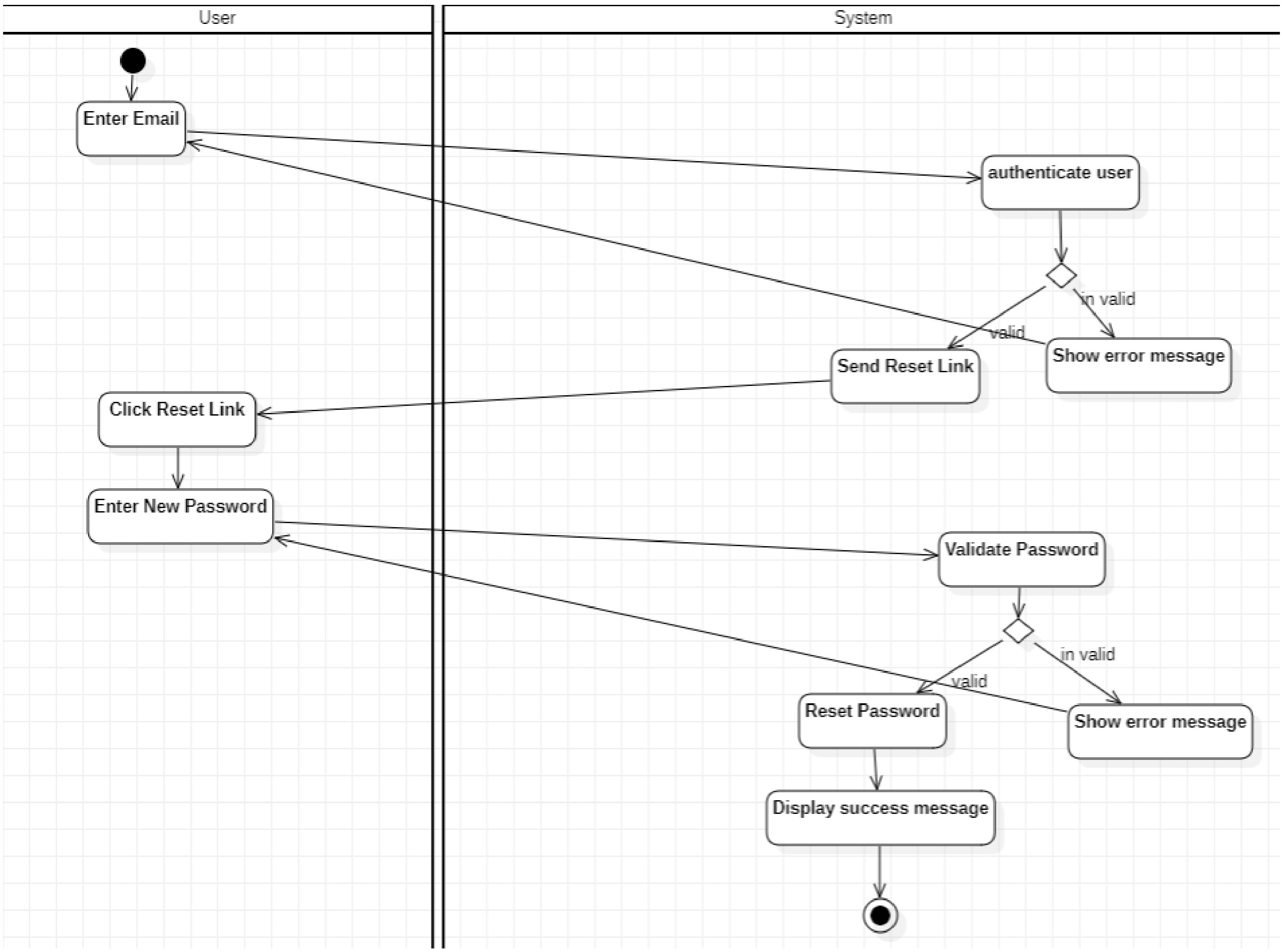
Sequence Diagram



1.4 Forgot Password

Use Case Title	Forget Password
Use Case ID	UC-004
Actor	**User**: A user who has forgotten their account password and wants to reset it.
Goal	To allow the user to securely reset their password using their registered email address.
Pre-conditions	1-The user must have a registered account with a valid email address. 2-The user must have access to the email account.
Post-conditions	1-The user receives a password reset link via email. 2-The user can successfully reset their password.
Exceptions	1-If the email sending fails due to a system error, notify the user. 2-If the reset link is expired or invalid, prompt the user to request a new link.
Alternative Flow	***Alternative Flow for Email Sending Failure**: • 5A. The system encounters an error while trying to send the email. • 5B. The system displays an error message and suggests the user try again later. ***Alternative Flow for Expired or Invalid Reset Link**: • 6A. The user clicks on an expired or invalid reset link. • 6B. The system displays an error message and prompts the user to request a new password reset link.
Main Flow	**User** **1-User Accesses Forgot Password Page**: **2-User Enters Email Address**: **3-Submit Request**: **4-User Checks Email**: **5-User Enters New Password**: **6-Submit New Password.** **System** **1-System Displays Forgot Password Page**: **2-System Validates Email**: **3-System Generates Password Reset Link** (If Email is Registered): **4-System Sends Password Reset Link**: **5-System Redirects to Password Reset Page**: **6-System Validates New Password**: **7-System Updates Password**: **8-System Sends Confirmation**:

Activity Diagram

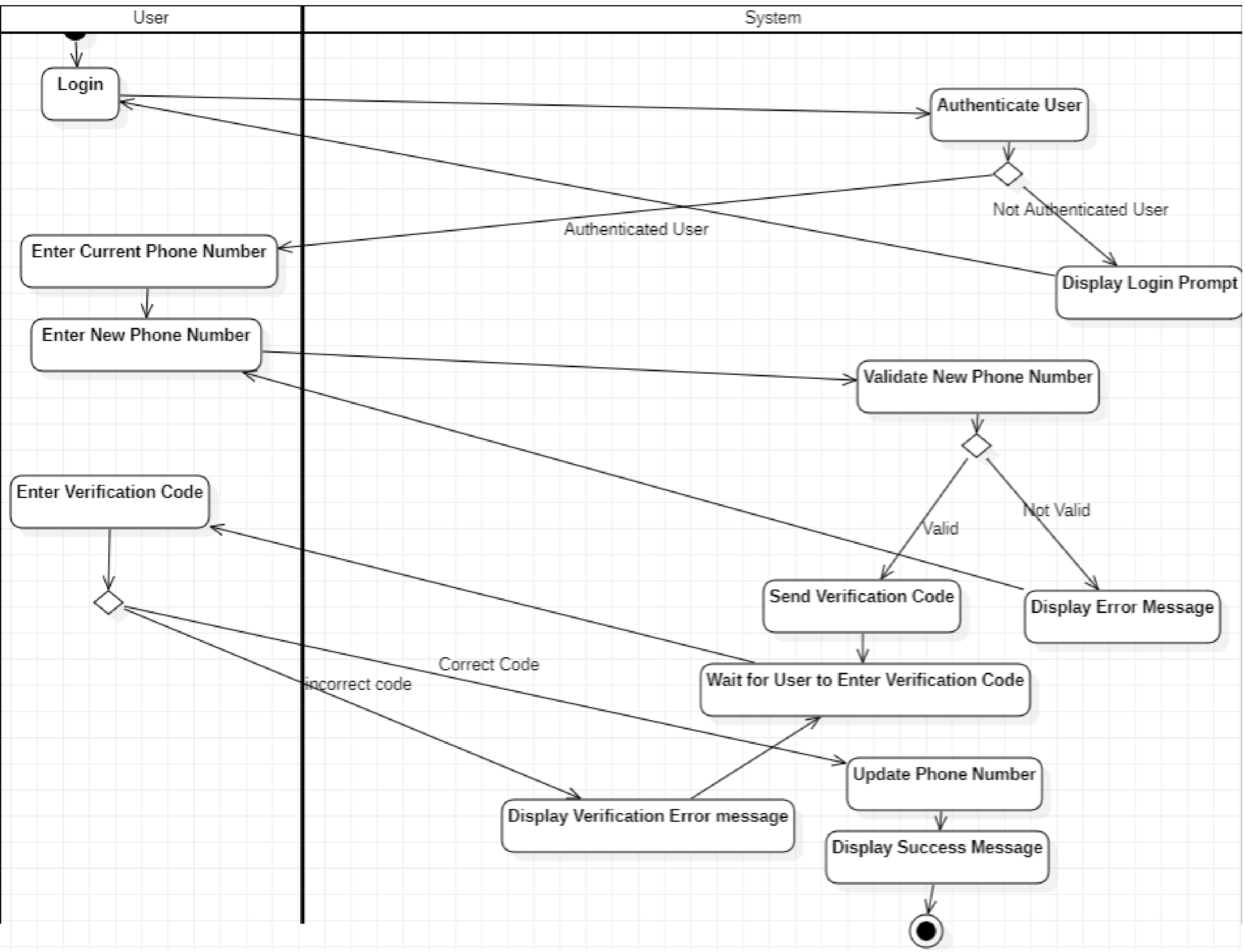


Sequence Diagram

1.5 Change Phone Number

Use Case Title	Change Phone Number	
Use Case ID	UC-005	
Actor	**User**: An authenticated user who wants to change their registered phone number.	
Goal	To enable the user to securely change their registered phone number.	
Pre-conditions	1- The user must be logged into their account. 2- The user must have access to their current phone number. 3- The user must have access to the new phone number they wish to use.	
Post-conditions	1- The user's phone number is successfully updated. 2- The user receives a confirmation notification via SMS to their new phone number.	
Exceptions	1- If the new phone number is already in use by another account, display an error message. 2- If the user fails to verify the new phone number within a specified time, prompt them to request a new verification code.	
Alternative Flow	***Alternative Flow for New Phone Number Already in Use**: • 6A. The system detects that the new phone number is already associated with another account. • 6B. The system displays an error message prompting the user to choose a different phone number. ***Alternative Flow for Verification Code Expiration**: • 9A. The user does not enter the verification code within the specified time frame. • 9B. The system displays a message indicating that the code has expired and provides an option to resend the verification code.	
Main Flow	User **1- User Accesses Change Phone Number Page.** **2- User Enters Current Phone Number** **3- User Enters New Phone Number.** **4- Submit Change Phone Number Form.** **5- User Receives SMS Code.** **6- User Enters Verification Code.**	System **1- System Displays Change Phone Number Page.** **2- System Validates Current Phone Number.** **3- System Validates New Phone Number.** **4- System Sends Verification Code.** **5- System Validates Verification Code.** **6- System Updates Phone Number.** **7- System Sends Confirmation.**

Activity Diagram



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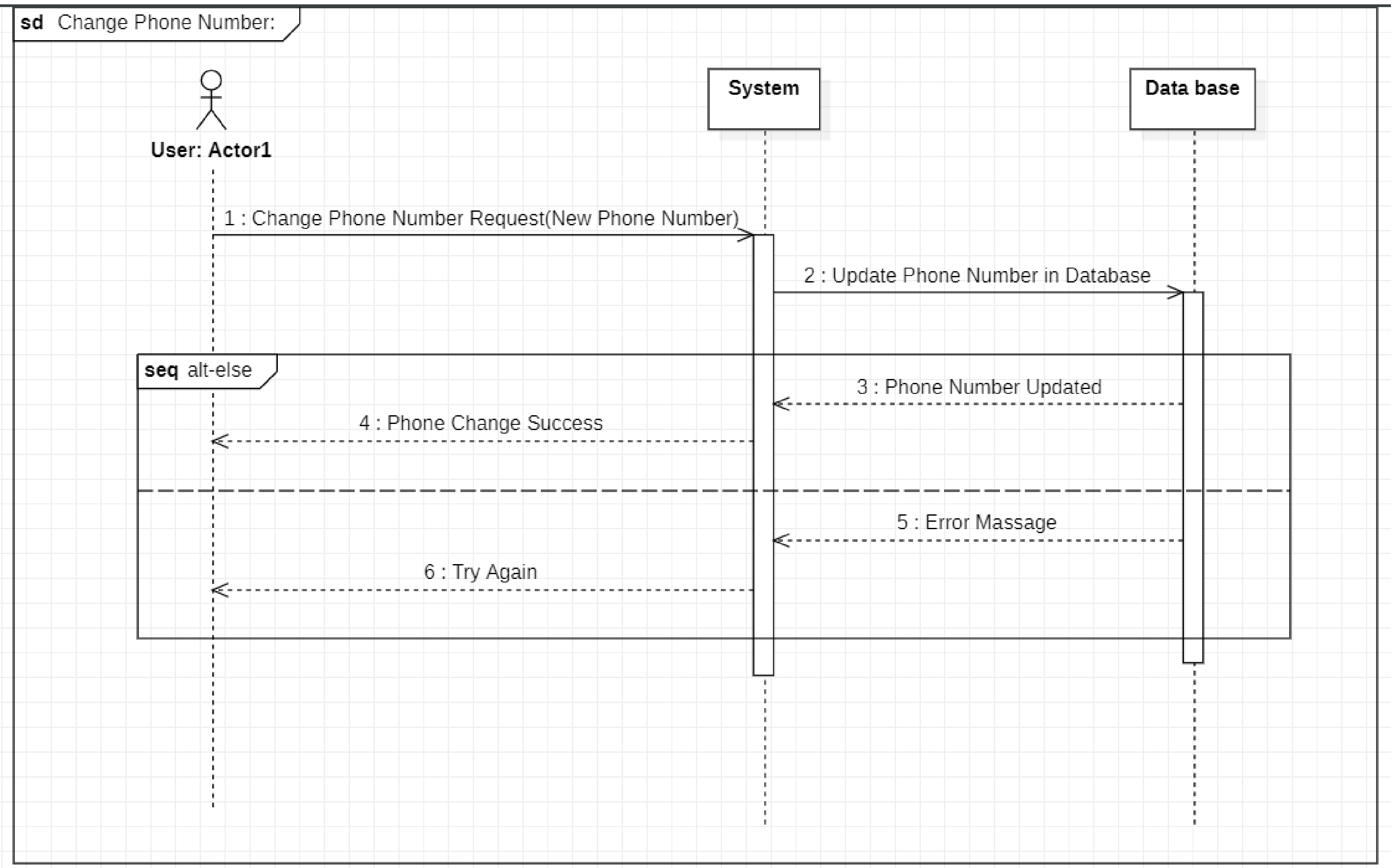
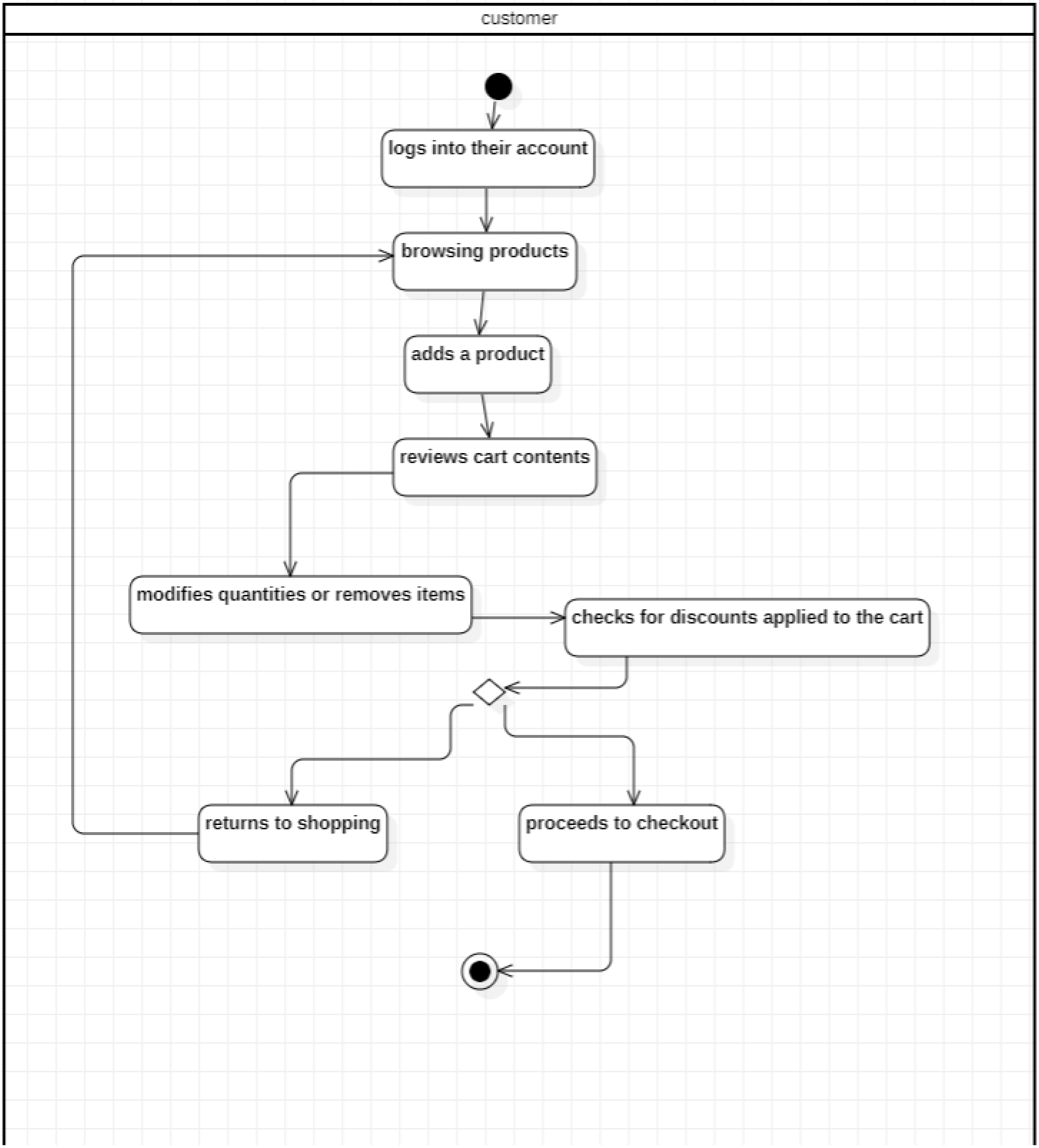


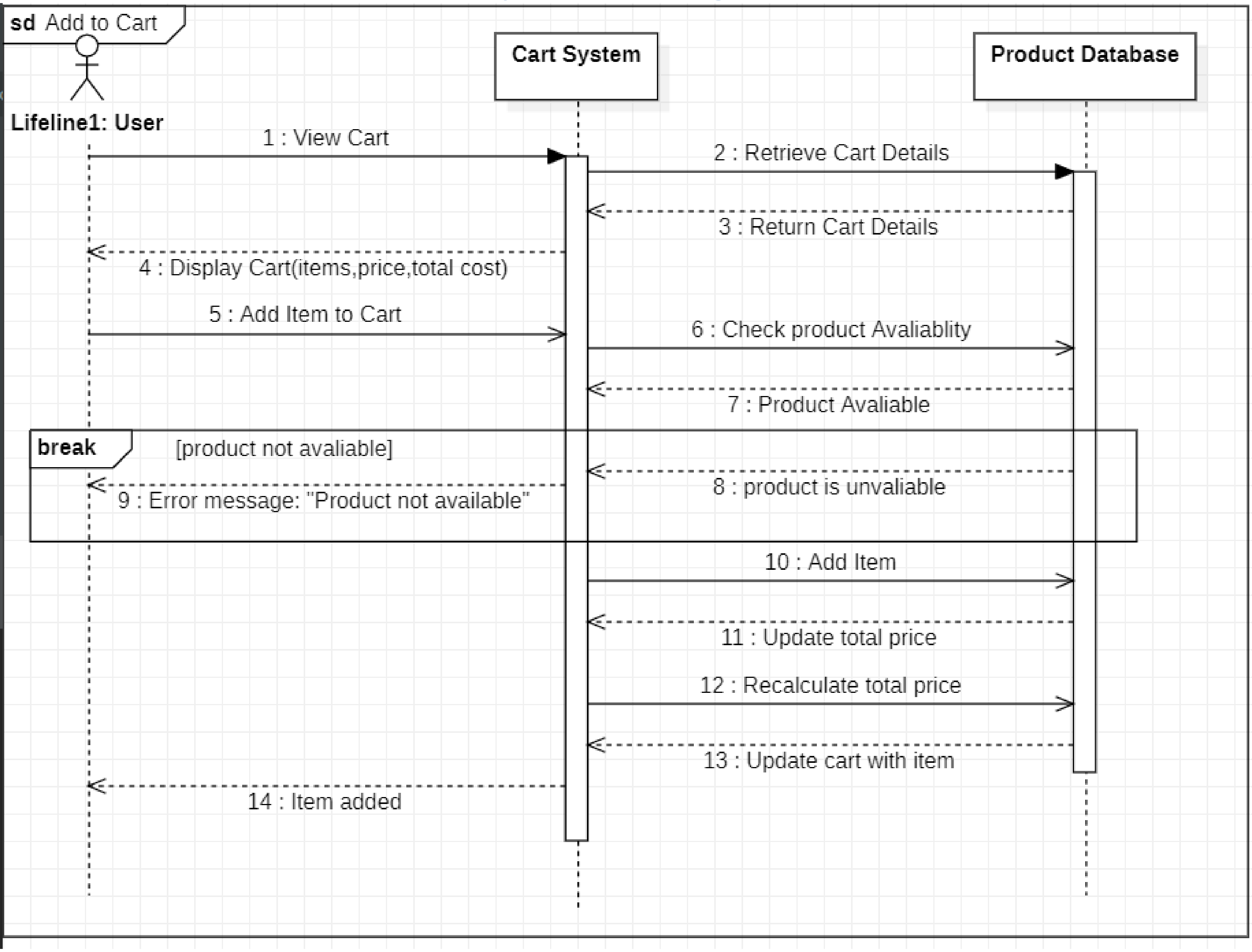
1.6 Manage Cart

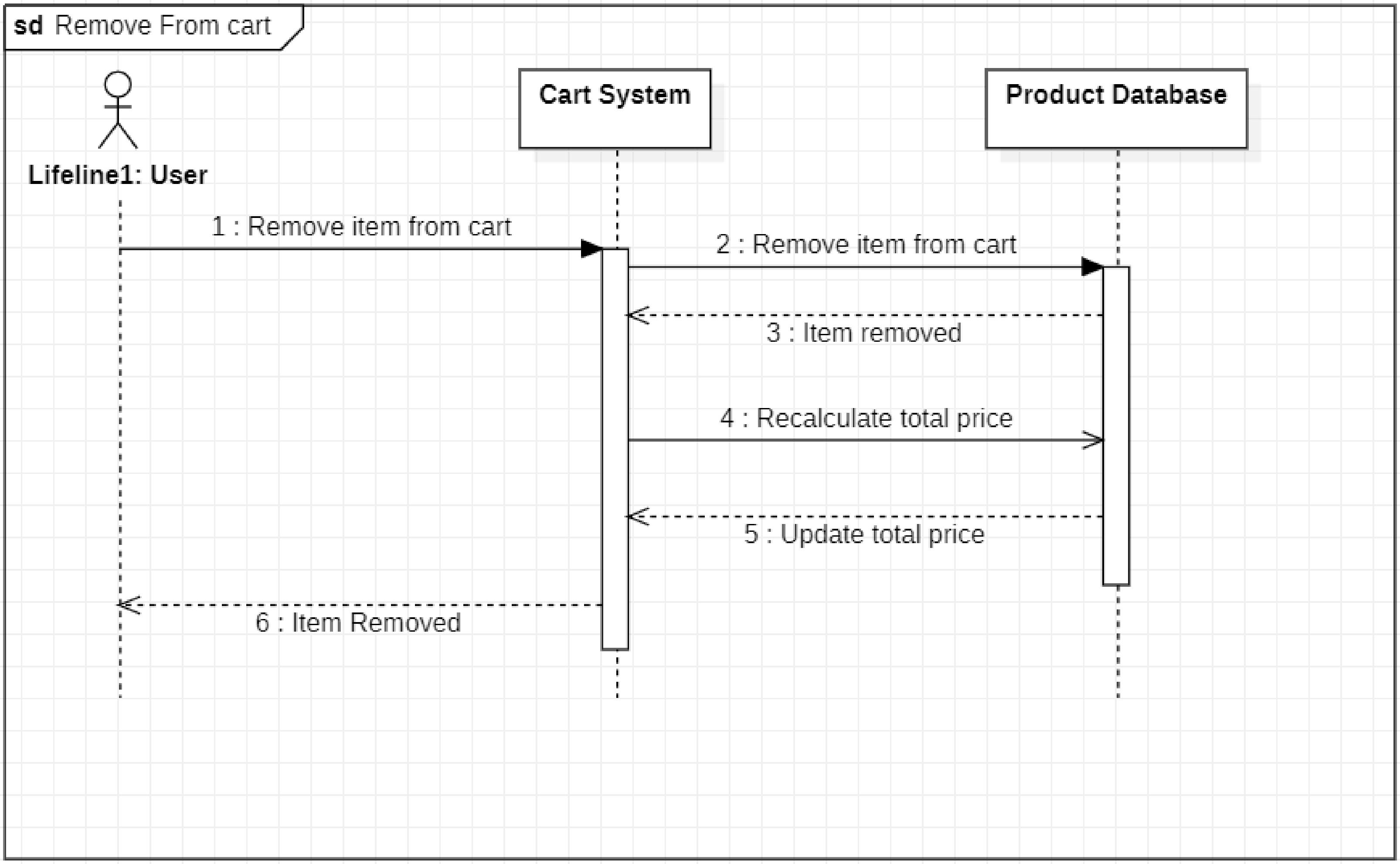
Title	Manage Cart	
Actors	customer who is shopping on the e-commerce site.	
Goal	To allow customers to add, remove, update quantities, and view items in their shopping cart	
Preconditions	1- The customer must be logged into their account . 2- The customer has items in their cart.	
Postconditions	The cart is updated based on the customer's actions (items added, removed, or quantities updated)	
Exceptions	1- **Out of Stock**: • If the customer attempts to add an item that is out of stock, the system displays an error message. 2- **Invalid Quantity**: • If the customer tries to update the quantity to a negative number or exceeds available stock, the system displays an error message.	
Alternative Flows	**Empty Cart**: If the cart is empty, the system displays a message indicating that the cart is empty and provides a link to continue shopping.	
Flow of Activity	**Actor**	**System**
	1- View Cart: The customer clicks on the Cart icon to view current items in the cart.	**1.1** The system displays the current items in the cart **1.2** showing details such as item names, quantities, prices, and total cost. **2.1** The system adds the item to the cart and **2.2** updates the total cost. **2.3** confirmation message displayed.
	2- Add Item to Cart: 1- The customer browses products and selects an item to add to the cart. 2- The customer clicks "Add to Cart."	**3.1** The system adds the item to the cart and **3.2** updates the total cost. **3.3** confirmation message displayed.
	3- Remove Item from Cart: 1-the customer identifies an item they want to remove. 2-The customer clicks "Remove" next to the item.	**4.1** The system updates the items **5.1** The system redirects the customer to the checkout process.
	4-Update Item. **5-Proceed to Checkout.**	

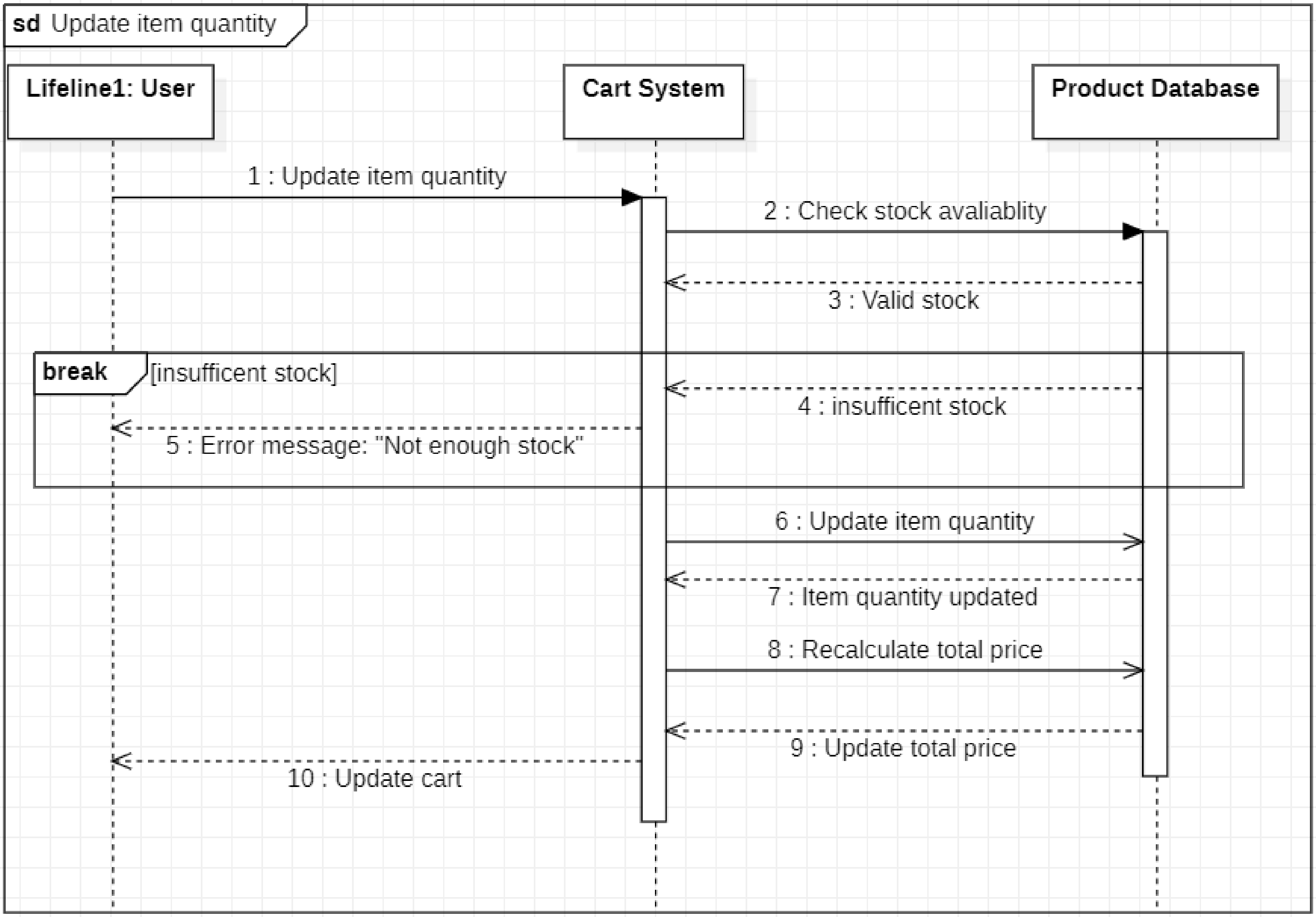
Activity Diagram



Sequence Diagram



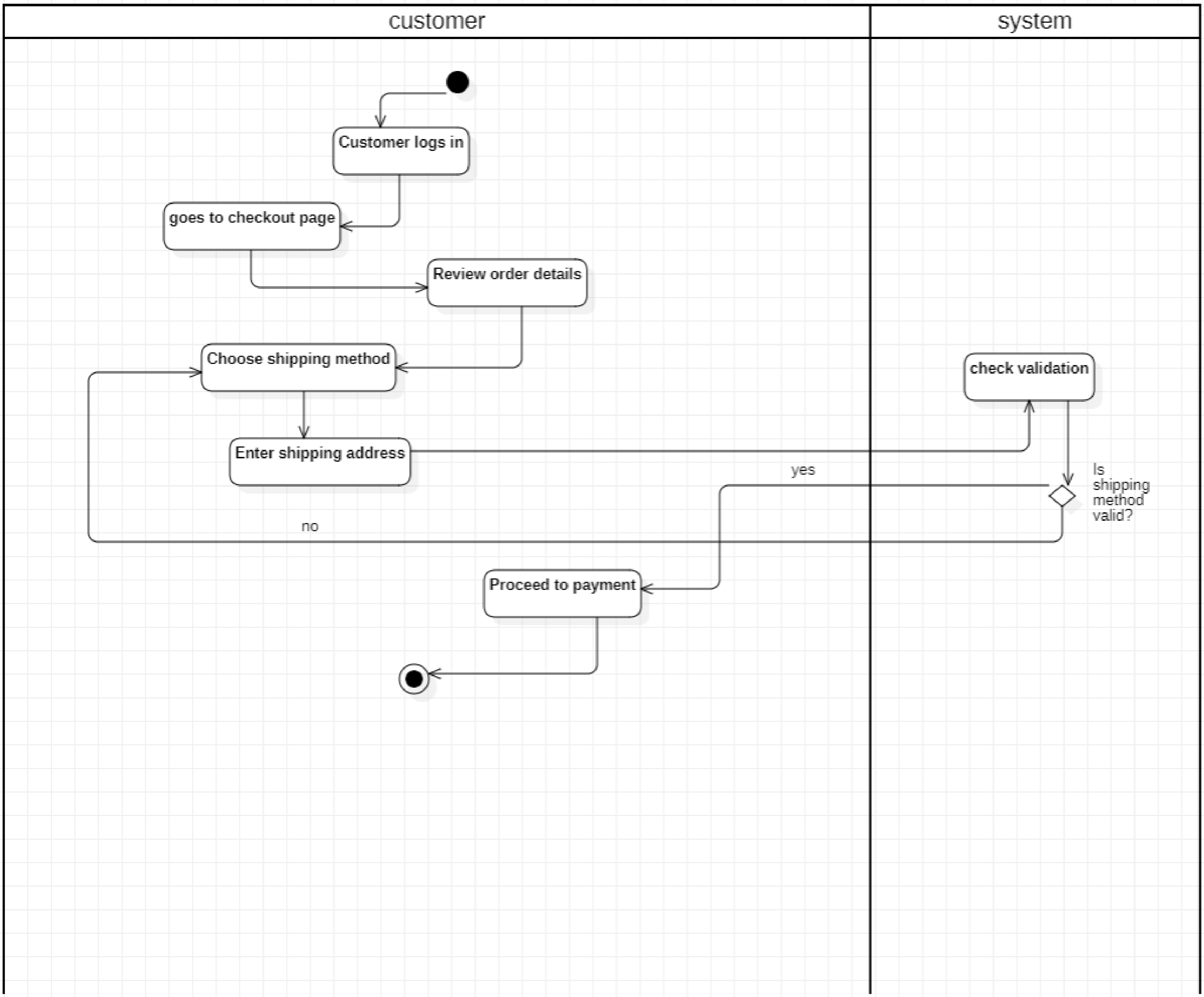

sd Remove From cart
Lifeline1: User
Cart System
Product Database
1 : Remove item from cart
2 : Remove item from cart
3 : Item removed
4 : Recalculate total price
5 : Update total price
6 : Item Removed


sd Update item quantity
Lifeline1: User
Cart System
Product Database
1 : Update item quantity
2 : Check stock avaliablity
3 : Valid stock
break
[insufficent stock]
4 : insufficent stock
5 : Error message: "Not enough stock"
6 : Update item quantity
7 : Item quantity updated
8 : Recalculate total price
9 : Update total price
10 : Update cart

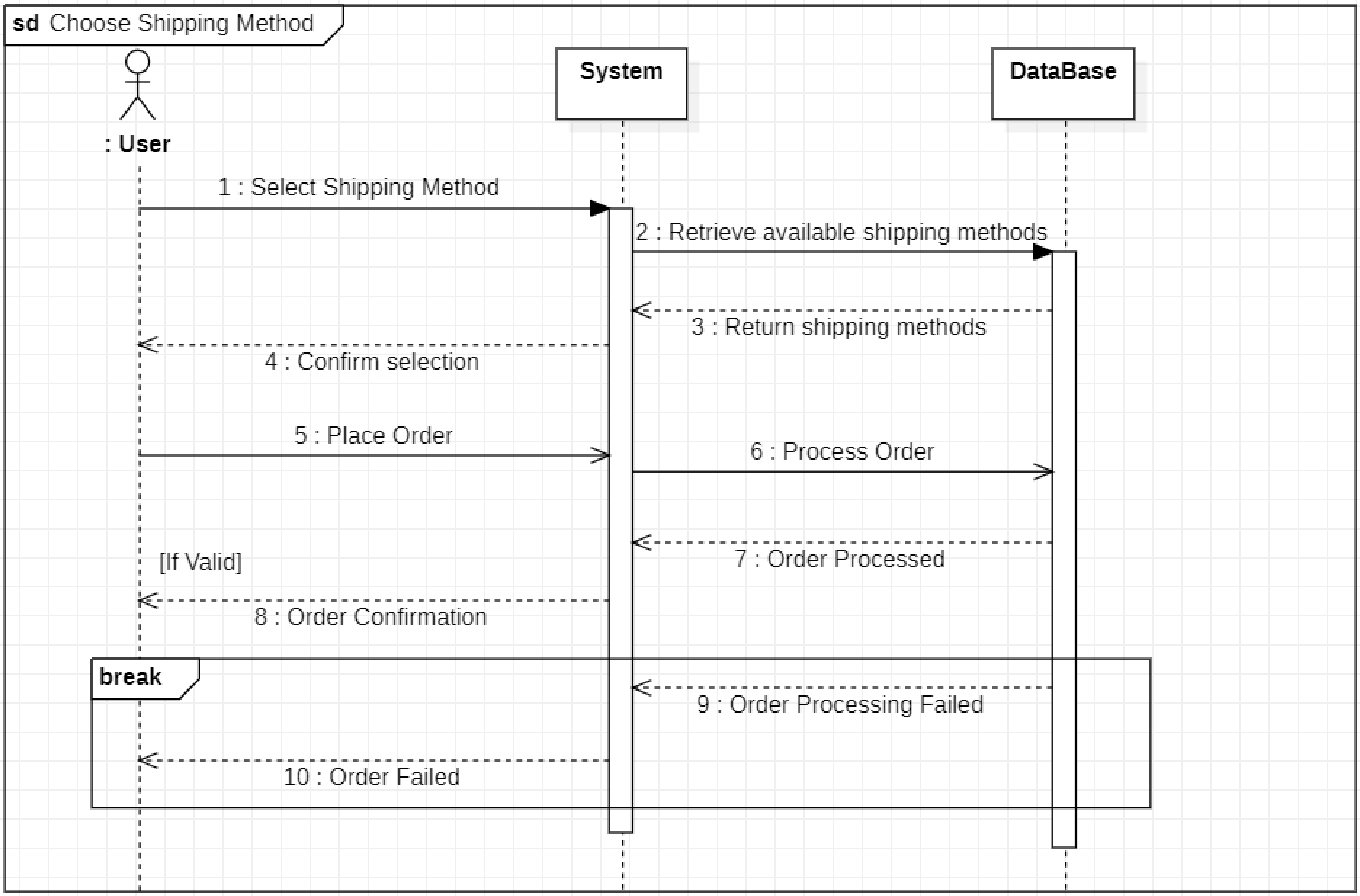
1.7 Choose Shipping Method

Title	Choose Shipping Method	
Actors	**Customer**: The user who is purchasing items on the e-commerce site.	
Goal	To allow customers to select their preferred shipping method during the checkout process.	
Preconditions	The customer has items in their cart and has initiated the checkout process.	
Postconditions	The selected shipping method is saved and used for order processing**.**	
Exceptions	**Error Retrieving Shipping Methods:** • If the system encounters an error when retrieving available shipping methods, it displays an error message and suggests trying again.	
Alternative Flows	**No Shipping Options Available:** If no shipping methods are available for the customer's location, the system displays a message indicating this and prompts the customer to contact support.	
Flow of Activity	**Actor** 1- **Access Shipping Options:** The customer reaches the "Shipping Method" section 2- **View Shipping Details:** Each shipping option includes relevant details 3- **Select Shipping Method**: 1-The customer reviews the available options and selects their preferred shipping method. 2-The customer clicks on the "Select" button next to their choice. **4-Confirm Selection**:	**System** 1.1The system displays available shipping methods **4.1**The system updates the order summary to reflect the chosen shipping

Activity Diagram



Sequence Diagram



1.8 Place an Order

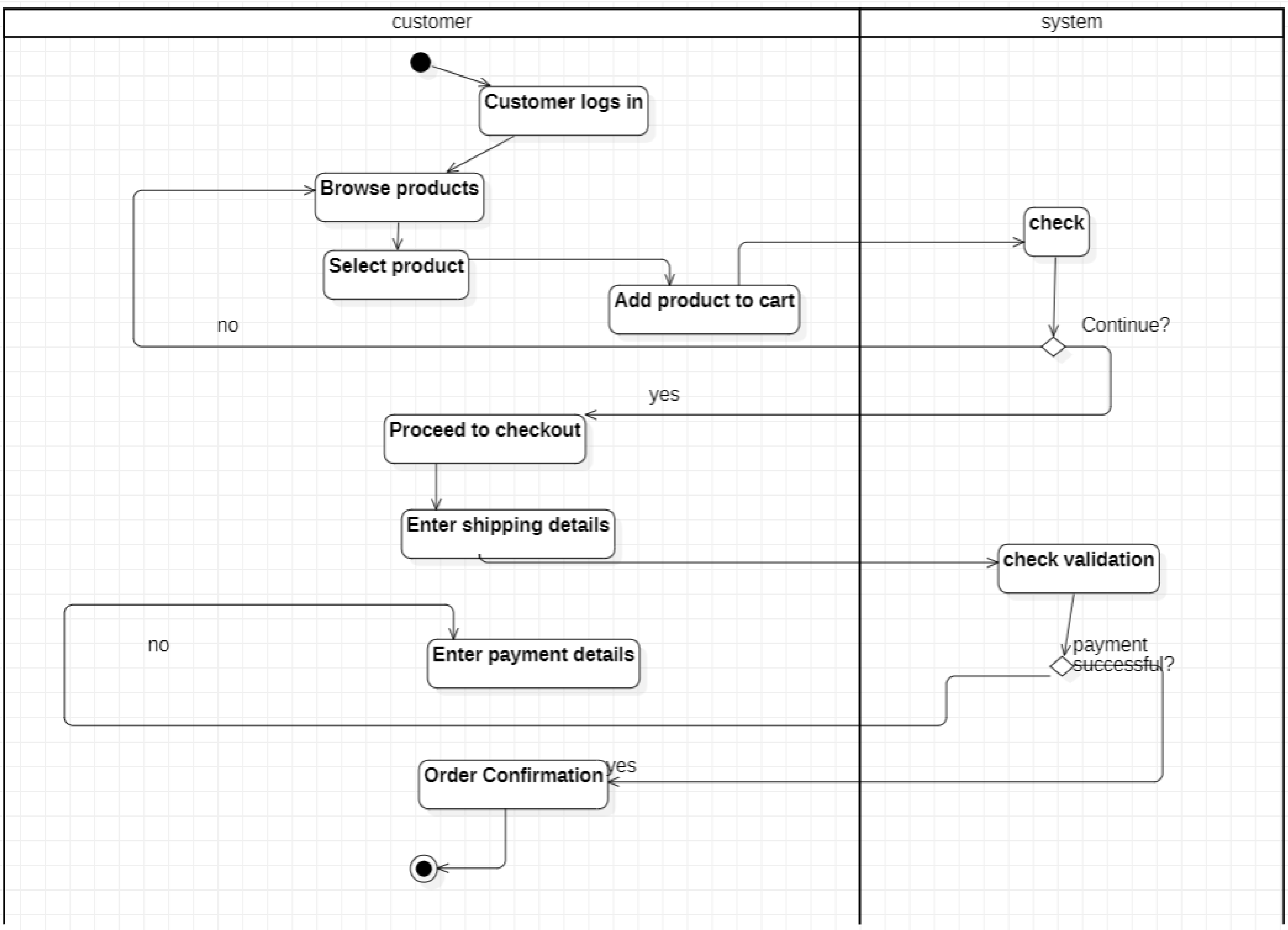
<table>
<tr><td>Title</td><td colspan="2">Place an Order</td></tr>
<tr><td>Actors</td><td colspan="2">Customer: The user who is completing a purchase on the e-commerce site.</td></tr>
<tr><td>Goal</td><td colspan="2">To allow customers to finalize their shopping experience by placing an order for items in their cart.</td></tr>
<tr><td>Preconditions</td><td colspan="2">1- The customer has items in their cart.
2- The customer has selected a shipping method.
3- The customer is logged into their account</td></tr>
<tr><td>Postconditions</td><td colspan="2">The order is placed successfully, and the customer receives confirmation of the order.</td></tr>
<tr><td>Exceptions</td><td colspan="2">1- Insufficient Stock:
If an item goes out of stock before the order is placed, the system informs the customer and may remove the item from the order.
2- Technical Issues:
If the system encounters an error while processing the order, it displays a generic error message and suggests trying again later.</td></tr>
<tr><td>Alternative Flows</td><td colspan="2">1- Payment Declined:
• If the payment is declined, the system displays an error message.
• The customer can choose to enter a different payment method or retry the current method.
2- Order Cancellation:
• After placing the order, if the customer wants to cancel it, they may be given the option to cancel the order within a specified timeframe.</td></tr>
<tr><td>Flow of Activity</td><td>Actor
1- Access Checkout:
The customer navigates to the checkout page after managing their cart.
2- Review Order Summary

3- Enter Payment Information:
The customer is prompted to enter payment details
4- Confirm Payment:
The customer reviews their payment details and clicks on "Place Order."
5- Order Confirmation:
6- End Transaction:</td><td>System
2.1The system displays an order summary

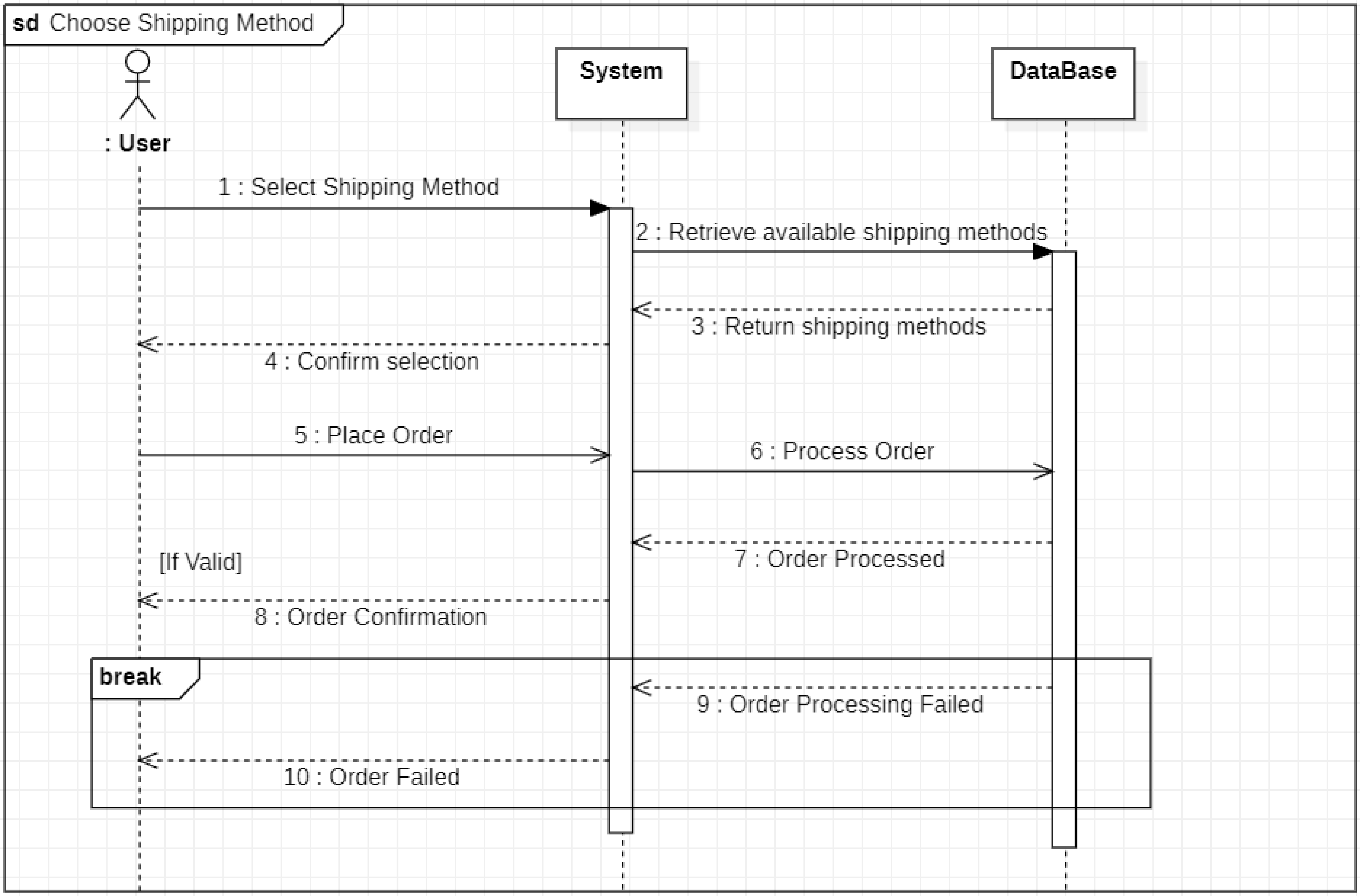
3.1The system may offer saved payment methods if the customer is logged in.

6.1The system updates inventory levels based on the items purchased</td></tr>
</table>

Activity Diagram



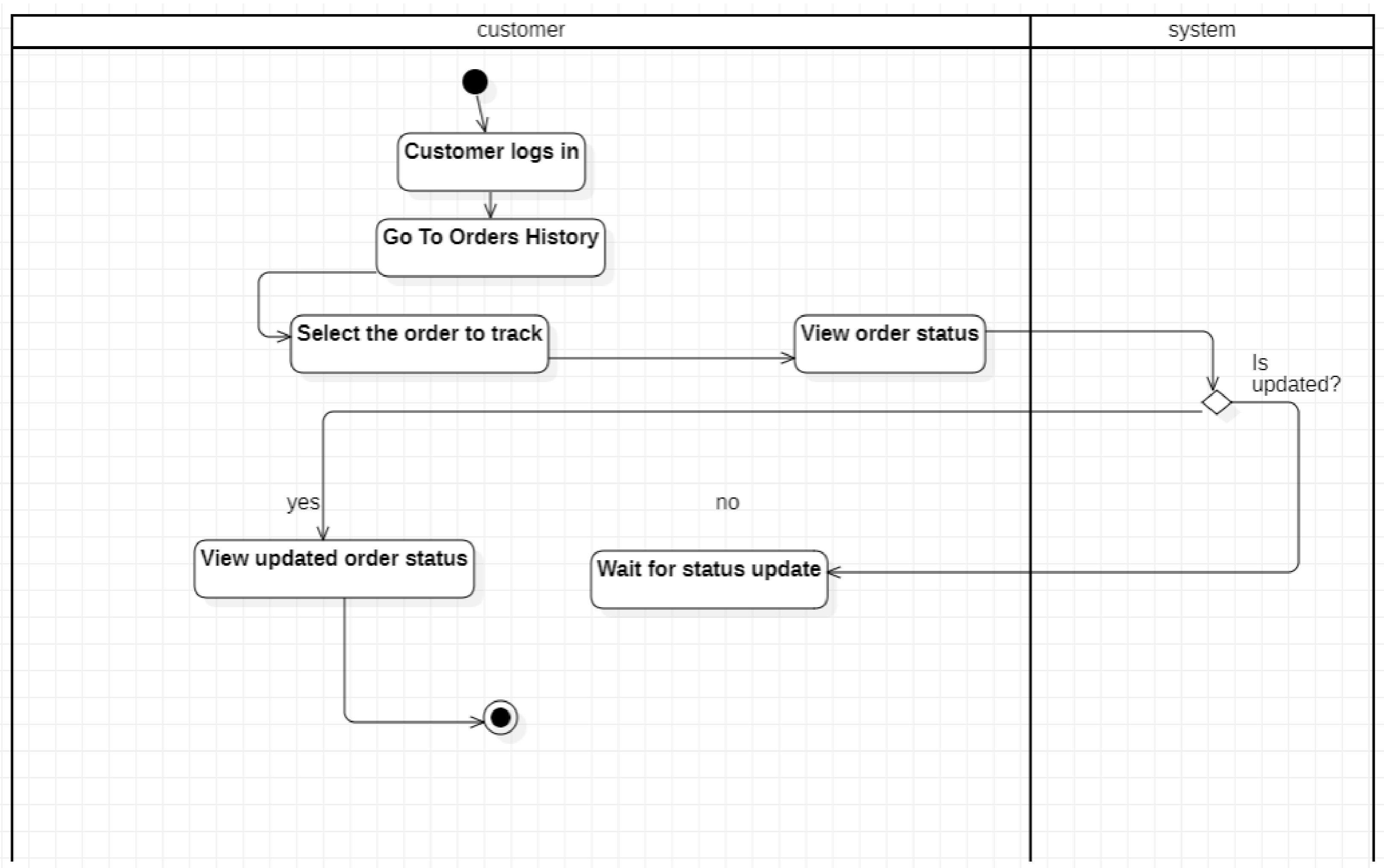
Sequence Diagram



1.9 Track Order Status

Title	Track Order Status	
Actors	**Customer**: The user who wishes to monitor the status of their placed orders.	
Goal	To enable customers to check the current status of their orders, including shipping and delivery information.	
Preconditions	1- The customer has placed at least one order. 2- The customer is logged into their account (if required).	
Postconditions	The customer receives the current status of their order, including any updates regarding shipping and delivery	
Exceptions	**Delayed Shipping**: • If there are unexpected delays in shipping, the system should notify the customer about the delay and provide updated estimates.	
Alternative Flows	**1- Order Not Found:** • If the customer enters an incorrect order number or selects an order that doesn't exist, the system displays an error message. **2- Technical Issues:** • If the system encounters an error retrieving order information, it displays a message prompting the customer to try again later.	
Flow of Activity	**Actor**	**System**
	1- The customer navigates to the "Track Order" section on their account page. **2- Select Order**: The customer selects the specific order they wish to track from a list of past orders. **3- View Order Status**: **4- Receive Notifications** (optional): The customer may opt to receive notifications via email updates on their order status.	**3.1** The system displays the current status of the selected order.

Activity Diagram



Sequence Diagram

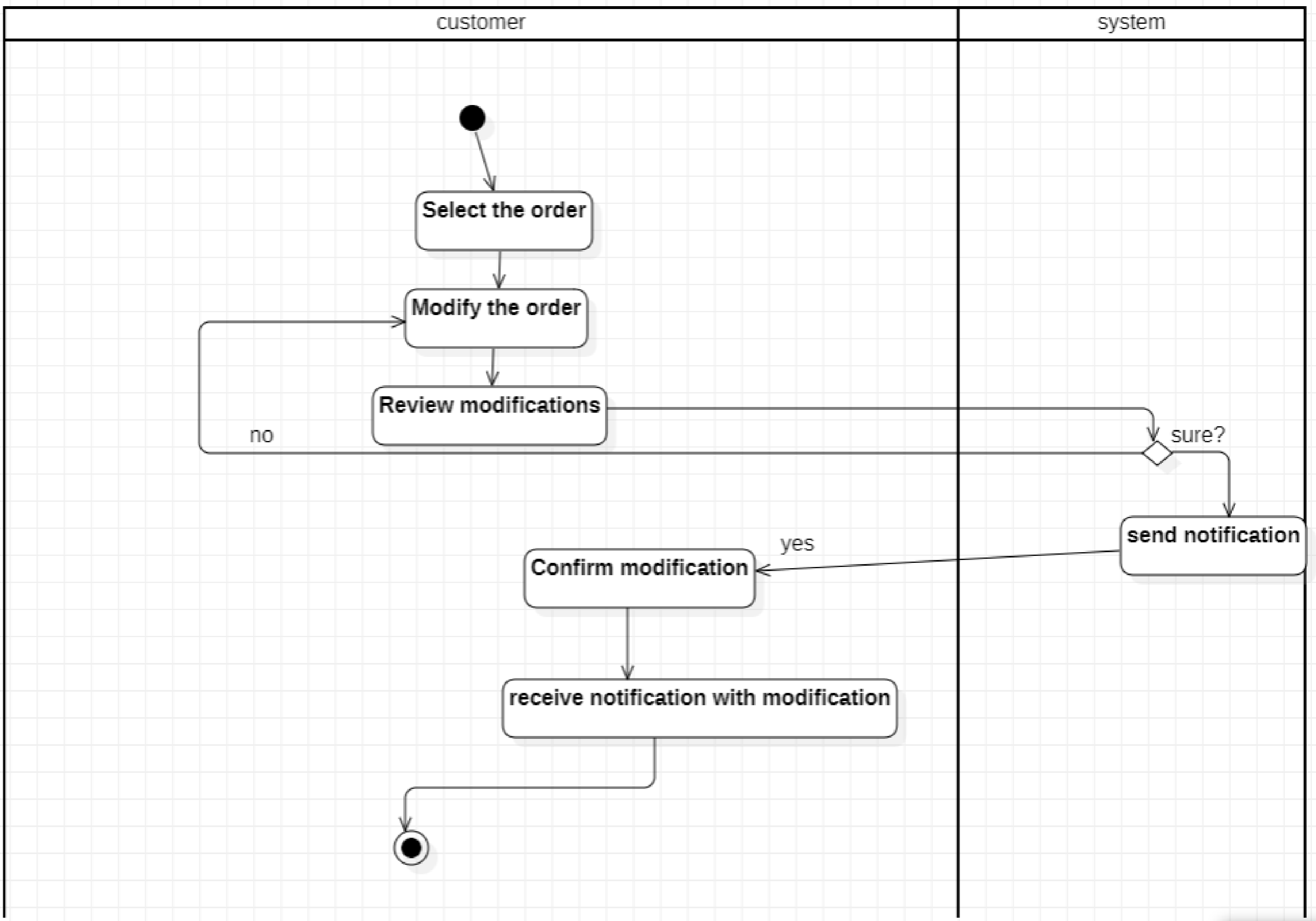
1.10 Modify an Order

<table>
<tr><td>Title</td><td colspan="2">Modify an Orde</td></tr>
<tr><td>Actors</td><td colspan="2">Customer: The user who wishes to change details of an existing order.</td></tr>
<tr><td>Goal</td><td colspan="2">To allow customers to make changes to their order, before the order is finalized or shipped.</td></tr>
<tr><td>Preconditions</td><td colspan="2">The customer has placed an order that is eligible for modification (the order has not yet been shipped).</td></tr>
<tr><td>Postconditions</td><td colspan="2">The order is updated according to the customer's changes, and the customer receives confirmation of the modifications.</td></tr>
<tr><td>Exceptions</td><td colspan="2">1- Insufficient Stock:
-If the customer attempts to increase the quantity of an item that is out of stock, the system notifies them and may suggest alternatives.
2-Payment Issues:
-If adding new items requires additional payment, the system prompts the customer to enter payment details.</td></tr>
<tr><td>Alternative Flows</td><td colspan="2">1- Modification Not Allowed:
• If the order is already in a state that doesn’t allow modifications (e.g., shipped or delivered), the system displays a message indicating that modifications are not possible.
2- Error in Modification Process:
• If the system encounters an error while processing the modifications, it informs the customer and allows them to retry.</td></tr>
<tr><td>Flow of Activity</td><td>Actor
1- Access Order History:
The customer logs into their account and navigates to the "Order History"
2- Select Order to Modify:
The customer selects the specific order they wish to modify from the list of past orders.
3- Initiate Modification:</td><td>System
4.1 The system displays the order details, allowing the customer to change

5.1 The system shows an updated order summary, including any changes made.
6.1 The system processes the changes and updates the order in the database</td></tr>
</table>

	The customer clicks on the "Modify Order" button 4- **Make Changes:** -The customer makes the desired changes. 5- **Review Changes** 6- **Confirm Modification:** -The customer clicks on the "Confirm Changes" button.	**7.**1The system provides a confirmation message detailing the modifications**.**

Activity Diagram



Sequence Diagram

1.11 Cancel an Order

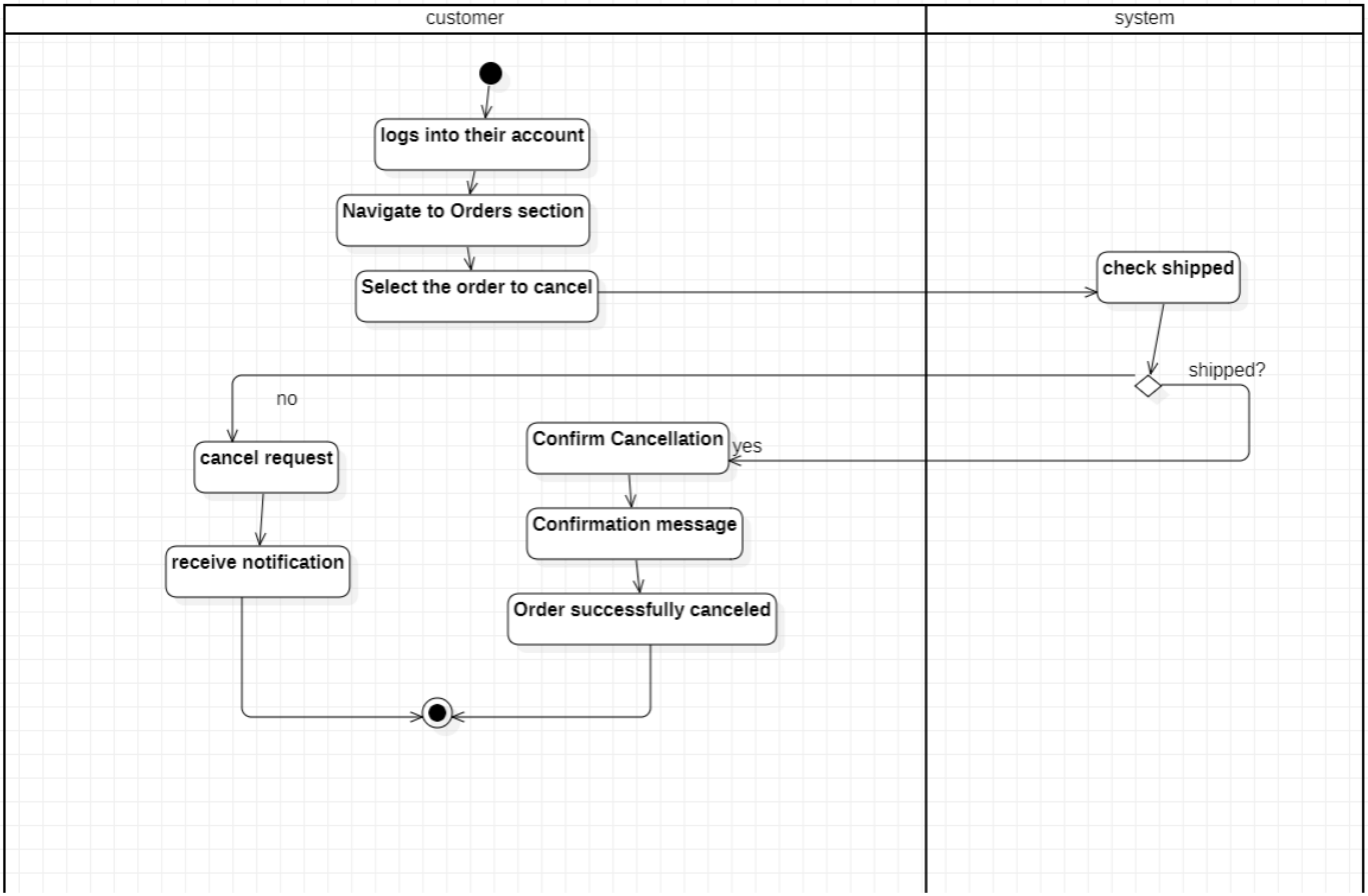
<table>
<tr><td>Title</td><td colspan="2">Cancel an Order</td></tr>
<tr><td>Actors</td><td colspan="2">Customer: The user who wishes to cancel a previously placed order.</td></tr>
<tr><td>Goal</td><td colspan="2">To allow customers to cancel their orders before they are processed, shipped, or delivered.</td></tr>
<tr><td>Preconditions</td><td colspan="2">1- The customer has placed an order.
2- The order is still in a cancellable state (not yet shipped or delivered).</td></tr>
<tr><td>Postconditions</td><td colspan="2">The order is canceled, and the customer receives confirmation of the cancellation.</td></tr>
<tr><td>Exceptions</td><td colspan="2">1- Payment Issues:
• If the payment for the order was not successfully processed (e.g., declined card), the system notifies the customer and may not allow cancellation.
2- System Maintenance:
• If the system is undergoing maintenance or is temporarily unavailable, the customer is informed and advised to try again later.</td></tr>
<tr><td>Alternative Flows</td><td colspan="2">1- Order Not Cancellable:
• If the order has already been shipped or delivered, the system informs the customer that the order cannot be canceled and may provide options for returns instead.
2- Error in Cancellation Process:
• If there’s an error while processing the cancellation, the system displays a message informing the customer and allows them to retry.</td></tr>
<tr><td>Flow of Activity</td><td>Actor
1- Access Order History:
The customer logs into their account and navigates to the "Order History"

2- Select Order to Cancel:
The customer selects the specific order they wish to cancel from the list of past orders.
3- Initiate Cancellation:</td><td>System
4.1 The system prompts the customer to confirm the cancellation, warning them about the implications.

5.1 The system updates the order status to "Canceled" in the database.
5.2 If applicable, the system initiates the refund process</td></tr>
</table>

	The customer clicks on the "Cancel Order" button **4- Confirm Cancellation**: The customer confirms their decision to cancel. **5- Process Cancellation**	**6.1** The system displays a cancellation confirmation message.

Activity Diagram

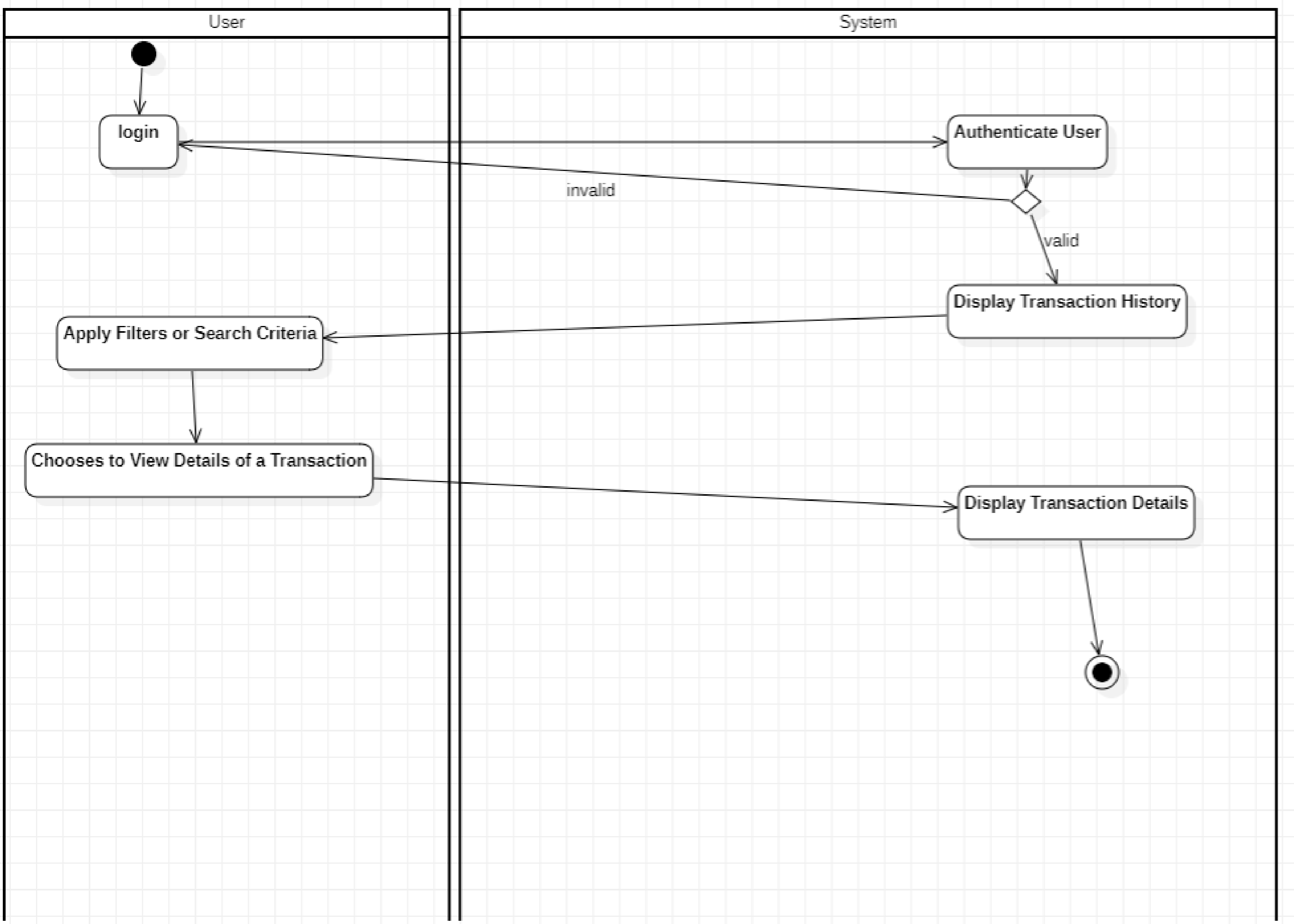


Sequence Diagram

1.12 View Transactions

Title	View Transaction	
Actors	Customer	
Goal	Enable customers to view past and recent transactions on the website.	
Preconditions	1. The customer is logged into their account. 2. The customer has a transaction history.	
Postconditions	The customer successfully views past and recent transactions.	
Exceptions	1. **System Error:** - If there's an issue retrieving transactions, the system displays an error message.	
Flow of Activity	**Actor**	**System**
	1- Log In: The customer logs into their account.	**1.1** The system authenticates the customer. **1.2** Provides access to their account.
	2- Navigate to Transactions: The customer clicks on the "Transactions" section.	**2.1** showing details such as item names, quantities, prices, and total cost.
	3- View Transaction: The customer reviews the detailed transaction information.	**3.1** The system displays transaction details, including order date, purchased items, total amount, payment status, and shipping details.
	4- Download Receipt : The customer clicks "Download Receipt" for a specific transaction (if needed).	**4.1** The system generates and downloads the receipt for the selected transaction.

Activity Diagram



Sequence Diagram

2.1.4 Non-Functional Requirements

- ***Usability:***
  The application should be user-friendly and intuitive for both caregivers and doctors to use.

- ***Reliability:***
  Having a good backend using Firebase we will have 99.95% uptime.

- ***Maintainability:***
  Our team will consistently test the app and fix any bugs (if found) ASAP.

- ***Scalability:***
  The App will be developed by adding new features.

- ***Security:***
  By using Firebase security metrics (App check) the data of our user "doctors or caregivers and their chats" should be protected from any malicious attack and protect patient privacy by complying with medical regulations and standards.

- ***Performance:***
  The project should have a fast response time and run smoothly & efficiently.

- ***Compatibility:***
  The application should be working on old and new android (Android 9.0+) and IOS smartphones (IOS 15.0+)